

Curriculum Vitae

Mingsong Li

School of Earth and Space Sciences, Peking University
No. 5 Yiheyuan Rd, Haidian District, Beijing, China 100871

Website: mingsongli.com

E-mail: msli@pku.edu.cn; limingsonglms@gmail.com

Research Interests: Paleoclimatology; Paleohydrology; Sedimentology; Earth System Modeling; Data Assimilation; Time-Series Analysis; Software Development

1. Education

2016 Ph.D., China University of Geosciences (Wuhan), Geology
2014–2016 Johns Hopkins University (Visiting Ph.D. Student, CSC Fellow)
2012 M.S., Jilin University, China, Paleontology and Stratigraphy
2009 B.S., Jilin University, China, Geology

2. Professional Experience

Assistant Professor, Peking University (2020–present)
Assistant Research Professor, Pennsylvania State University (2019–2020)
Postdoctoral Researcher, Pennsylvania State University (2017–2018)
Postdoctoral Researcher, George Mason University (2016–2017)
Research Assistant, School of Earth Sciences, China University of Geosciences (2012–2016)
Research Assistant, College of Earth Sciences, Jilin University (2009–2012)

3. Selected Awards and Honors

2026 Sun Shu Prize, International Association of Sedimentologists (IAS)
2025 The 4th “China Sun Shu Award”
2025 Stanford/Elsevier World's Top 2% Scientists
2025 China's Top 10 Advances in Paleontology (2024)
2024 Stanford/Elsevier World's Top 2% Scientists
2024 Outstanding Editorial Group Member, National Science Review
2021 National Young Talents Program, China
2021 Peking University Boya Young Fellow
2021 AGU's Outstanding Reviewers
2019 Top 1% Peer Reviewer Awards, Publons, Web of Science Group

4. Publications

Google Scholar: <https://scholar.google.com/citations?user=NYUXDW0AAAAJ>

Oct. 1, 2026: h-index: 29; citations: 4,600

* = Corresponding author; # = Student advisee

In revision or under review:

- # Kaixuan Ji, Xiaoyu Zhang, Zhixin Wang, Fanhao Gong, Huaichun Wu, Laiming Zhang, Ren Wei, Yuan Gao, Zhijun Jin, **Mingsong Li***. 2026. Chaotic Earth–Mars resonance drove sea level and climate–biodiversity dynamics in the latest Cretaceous. *Nature Communications*, in revision.
- # Jiahao Liu, Ruoyuan Qiu, Andy Ridgwell, Dan Zhu, Weiqi Yao, Liao Chang, Xinwen Zhang, Wang Miao, Ren Wei, Jianghui Du, Yongyun Hu, Zhijun Jin, **Mingsong Li***. 2026. Thermogenic carbon

- release reshaped the carbon cycle during the Paleocene–Eocene Thermal Maximum. *Nature Communications*, under review.
- # Zhixin Wang, Yang Liu, Feng Ding, Kaixuan Ji, Jiahao Liu, Ren Wei, Haotian Zhang, Hanyu Zhu, Zhijun Jin, **Mingsong Li***. 2026. Orbital modulation of nonlinear climate dynamics on early Mars. *Nature Astronomy*, under review.
- # Xinwen Zhang, Wanyi Lu, Babette Hoogakker, Weiqi Yao, Sifan Gu, Jiahao Liu, **Mingsong Li***. 2026. Reconstructing past bottom-water oxygen from benthic foraminiferal carbon-isotope offsets ($\Delta\delta^{13}\text{C}$): The BayDelC Bayesian framework. *Paleoceanography and Paleoclimatology*, under review.
- Mingsong Li***. 2026. Statistical tuning in cyclostratigraphy and astrochronology: foundations, validation, and methodological advances. *Earth-Science Reviews*, under review.
- Ren Wei, **Mingsong Li***, Xiaomei Wang*, Lin Dong, Yongyun Hu, Jian Cao, Meng Wang, Bo Jiu, Kaixuan Ji, Haotian Zhang, Zhijun Jin*. 2026. Orbitally gated monsoon wetlands as pulsed carbon sinks during the late Paleozoic Ice Age. *Proceedings of the National Academy of Sciences*, under review.
- Hongliang Lu, Baochun Huang, **Mingsong Li**, Qian Zhao, Xiang Li, Yilian Liao, Zhaowen Pei, Yi Xue, Josep M. Parés, Haotian Zhang, Qingqing Jiang. 2026. Synchronous End-Permian Mass Extinction Zenith in Marine and Low-Middle Latitude Land. *Proceedings of the National Academy of Sciences*, in revision.
- Tan Zhang, Chao Ma, Chao Li, Tailiang Fan, Chunju Huang, Anne-Christine Da Silva, Yifan Li, Qi Gao, He Huang, Meng Cheng, Yu Sui, **Mingsong Li**, Qiuyu Wang, Jiashan Lin, Hanting Zhong, Mingcai Hou, Chengshan Wang. 2026. Diachronous onset and comparatively clustered termination of the Ediacaran Shuram Excursion. *Nature Communications*, under review.

Published or accepted:

2026

85. Meng Wang, Yukun Shi, Aihua Yang, Haotian Zhang, Xiaoyu Zhang, Yujing Wu, Yujie Shi, Jianbo Liu, Ren Wei, Zhijun Jin, **Mingsong Li***. 2026. Coupled sea-level and biodiversity dynamics during the Cambrian SPICE event. *Nature Communications*, accepted.
84. Xiuwen Zhou, Ruiling Zhang, Man-Yin Tsang, **Mingsong Li**, Weiqi Yao. 2026. Coupled increases in pelagic fish productivity and export productivity during the Paleocene-Eocene Thermal Maximum. *Nature Communications*, 17(1): 10030. <https://doi.org/10.1038/s41467-026-77863-9>.
83. Gui-Mei Lu, Zhong-Shan Shen, Cheng-Long Deng, James G. Ogg, Ming-Dao Sun, **Mingsong Li**, Hua-Feng Qin, Ri-Xiang Zhu, Zhong-He Zhou, Yi-Gang Xu. 2026. Refined chronologies of magnetochron M0r reveal asynchronous terrestrial and marine carbon isotope responses to Oceanic Anoxic Event 1a. *Science Advances*, 12: eaea8374.
82. Qingqing Jiang, **Mingsong Li***, N. M. Papadomanolaki, Yonggang Liu, David De Vleeschouwer*. 2026. Precession-related variability in North Pacific deep water formation in the Early Eocene: Insights from cGENIE simulations. *Paleoceanography and Paleoclimatology*, 41: e2026PA005598.
81. Xiyue Liang, Dongjie Tang, Longfei Sun, Xiaoying Shi, **Mingsong Li**, Kunfeng Qiu, Baozeng Xie, Xinqiang Wang, Simon W. Poulton. 2026. Sunspot cycles drove decadal fluctuations in shallow water oxygenation in the Mesoproterozoic. *Earth and Planetary Science Letters*, 692: 120268.
80. Yilin Liu, Brandon B. Bell, Elizabeth M. Griffith, **Mingsong Li**, Yi Ge Zhang, Qingqing Jiang, Weiqi Yao. 2026. Decoupling of biological pump strength and efficiency in the eastern equatorial Pacific during the Middle Miocene Climate Optimum. *Earth and Planetary Science Letters*, 692: 120256.
79. Ren Wei, Zhijun Jin*, **Mingsong Li***, Yongyun Hu, Xiaomei Wang, Shuai Yuan, Meng Wang, Renda Huang, Haotian Zhang. 2026. Orbital modulation of meridional atmospheric heat transport and

- ITCZ variability during the Late Paleozoic Ice Age. *Journal of Geophysical Research: Atmospheres*, 131: e2026JD046636.
78. Zhijun Jin, Xiaomei Wang, Ren Wei, **Mingsong Li**, Yongyun Hu. **2026**. Multiscale hydrogeologic evolution and sedimentary responses during the Late Paleozoic Ice Age. *Earth-Science Reviews*, 282: 105675.
77. Renda Huang, Zhijun Jin, Ruoyuan Qiu, Xueyu Yao, Yinan Feng, Ren Wei, **Mingsong Li**, Youwei Wang. **2026**. Obliquity modulation pacing of mid–high-latitude lacustrine systems during the Late Paleozoic Ice Age. *Global and Planetary Change*, 264: 105554.
76. Ren Wei, Zhijun Jin*, **Mingsong Li***. **2026**. Geological records reveal chaotic diffusion during the Late Paleozoic Ice Age. *Fundamental Research*, <https://doi.org/10.1016/j.fmre.2026.03.023>.
75. Naihua Xue, Wei Wang, David De Vleeschouwer, Chengguo Guan, **Mingsong Li**, Meng Wang, Xunlai Yuan, Philippe Claeys. **2026**. Astrochronology of the Ediacaran Period reveals the temporal heterogeneity and sea-level pacing of Shuram excursion. *Precambrian Research*, 445: 108222.
74. Frits Hilgen, Jacques Laskar, **Mingsong Li**, Thomas Westerhold. **2026**. Cyclostratigraphy and astrochronology: The ultimate timekeepers of Earth’s history. *PAGES Magazine*, 34(1): 16–17.
73. **Mingsong Li**, Weijie Zhao. **2026**. Exploring the chaotic future and the deterministic past of the Solar System: An interview with Jacques Laskar. *National Science Review*, 13(4): nwag002. <https://doi.org/10.1093/nsr/nwag002>.
72. **Mingsong Li***. **2026**. Acycle: Enhanced time-series analysis software for geoscience research and education. *Science China Earth Sciences*, 69(3): 1066–1071. <https://doi.org/10.1007/s11430-025-1848-7>.
- Mingsong Li***. **2026**. Acycle: Enhanced time-series analysis software for geoscience research and education. *Scientia Sinica Terrae*, 56(3): 1100–1105. <https://doi.org/10.1360/N072025-0527>.
- # 71. Haotian Zhang, Yan Chen, Paul B. Wignall, Kaixuan Ji*, Meng Wang, Xiaoyu Zhang, Hanyu Zhu, Ren Wei, Zhijun Jin, **Mingsong Li***. **2026**. Orbital forcing of paleoclimate and sedimentation in the Early Triassic: Insights from the Dayulin Section, North China. *Science China Earth Sciences*, 69(9): 3306–3320. <https://doi.org/10.1007/s11430-026-1953-3>.
- Haotian Zhang, Yan Chen, Paul B. Wignall, Kaixuan Ji*, Meng Wang, Xiaoyu Zhang, Hanyu Zhu, Ren Wei, Zhijun Jin, **Mingsong Li***. **2026**. Orbital forcing of paleoclimate and sedimentation in the Early Triassic: Insights from the Dayulin Section, North China. *Scientia Sinica Terrae*, 56(9): 3355–3370. <https://doi.org/10.1360/N072026-0067>.
- # 70. Qian He, Haotian Zhang, Stephen P. Hesselbo, Sofie Lindström, Sev Kender, Christian J. Bjerrum, **Mingsong Li***, Jianxin Yu, Yongdong Wang, Thomas J. Algeo, Yong Lei, Jun Shen*. **2026**. Astronomically tuned chronostratigraphy of an Upper Triassic–Middle Jurassic lacustrine succession in Southwest China. *Global and Planetary Change*, 259: 105325. <https://doi.org/10.1016/j.gloplacha.2026.105325>.
69. Ke Xu, Nan Wu, Jianye Ren, David B. Kemp, Lin Ye, Zhongheng Sun, Xiangrong Yang, Xianghui Zhang, **Mingsong Li**. **2026**. Orbital forcing across a predominantly eolian record of the Eocene–Oligocene transition, Inner Mongolia, China. *Global and Planetary Change*, 257: 105247. <https://doi.org/10.1016/j.gloplacha.2025.105247>.

2025

- # 68. Haotian Zhang, Yan Chen, James G. Ogg, Zhiming Sun, Paul B. Wignall, Meng Wang, Haoxun Zhang, Xiaoyu Zhang, Yang Zhang, Keke Huang, Hongliang Lu, Baochun Huang, **Mingsong Li***. **2025**. Astronomically calibrated integrated stratigraphy of the Induan Stage (Early Triassic) and significance for the Permian–Triassic mass extinction and aftermath. *Earth and Planetary Science Letters*, 669: 119563.

- # 67. Qingqing Jiang, **Mingsong Li***, Weiqi Yao, Ren Wei, Kaixuan Ji, Haotian Zhang, Zhijun Jin. **2025**. Astrochronology of the Paleocene–Eocene Thermal Maximum on the East Tasman Plateau. *Global and Planetary Change*, 252: 104882. <https://doi.org/10.1016/j.gloplacha.2025.104882>.
- 66. Ren Wei, Zhijun Jin, **Mingsong Li**, Shuai Yuan, Yongyun Hu, Lin Dong, Rui Zhang, Juye Shi. **2025**. Tectonic–astronomical interactions in shaping late Paleozoic climate and organic carbon burial. *Nature Communications*, 16: 8805.
- 65. Tan Zhang, Chao Ma, Yifan Li, Chao Li, Anne-Christine Da Silva, Tailiang Fan, Qi Gao, Mingzhi Kuang, Wangwei Liu, **Mingsong Li**, Mingcai Hou. **2025**. Astronomically calibrating early Ediacaran evolution. *Nature Communications*, 16: 3049.
- 64. Renda Huang, **Mingsong Li**, Ruoyuan Qiu, Youwei Wang, Rui Zhang, Zhijun Jin. **2025**. Enhancing astronomical signal detection in cyclostratigraphy by superimposed averaging of paleoclimate proxies. *Paleoceanography and Paleoclimatology*, 40: e2025PA005135.
- # 63. Haoxun Zhang, **Mingsong Li***. **2025**. Reconstruction of global sea surface temperatures during the Early Eocene Climatic Optimum using paleoclimate data assimilation. *Geological Journal of China Universities*, 31(3): 275–285.
- # 62. Haoxun Zhang, **Mingsong Li***, Yongyun Hu. **2025**. Paleoclimate data assimilation: Principles and prospects. *Science China Earth Sciences*, 68(2): 407–424. <https://doi.org/10.1007/s11430-024-1439-y>.
- Haoxun Zhang, **Mingsong Li***, Yongyun Hu. **2025**. Paleoclimate data assimilation: Principles and prospects. *Scientia Sinica Terrae*, <https://doi.org/10.1360/SSTe-2024-0090>.
- 61. Kai Wang, **Mingsong Li**, Peng Tang, Junxuan Fan, Renbin Zhan, Jianbo Liu. **2025**. High-resolution sedimentary cyclostratigraphy and astronomical signals in the Upper Ordovician of Southwest China. *Palaeoworld*, 34(3): 100893. <https://doi.org/10.1016/j.palwor.2024.11.001>.

2024

- 60. **Mingsong Li***, Lee R. Kump, Andy Ridgwell, Jessica E. Tierney, Gregory J. Hakim, Steven B. Malevich, Christopher J. Poulsen, Robert Tardif, Haoxun Zhang, Jiang Zhu. **2024**. Coupled decline in ocean pH and carbonate saturation during the Palaeocene–Eocene Thermal Maximum. *Nature Geoscience*, 17: 1299–1305.
- 59. **Mingsong Li***, Weiqi Yao, Jun Shen, Yuangeng Huang, Yi Yang. **2024**. Orbital-scale spatiotemporal dynamics of ocean deoxygenation during Paleocene–Eocene Thermal Maximum. *China Basic Science*, 26: 24–30.
- # 58. Meng Wang, **Mingsong Li***, Elizabeth A. Hajek, David B. Kemp, Yujing Wu, Hanyu Zhu, Chunju Huang, Haotian Zhang, Kaixuan Ji, Rui Zhang, Ren Wei, Zhijun Jin*. **2024**. Assessing the preservation of orbital signals across different sedimentary environments: Insights from stochastic sedimentation modeling. *Earth and Planetary Science Letters*, 642: 118866.
- 57. Kai Wang, **Mingsong Li**, Aonan Zheng, Rongchang Wu, Xin Wei, Jianbo Liu. **2024**. Comprehensive cyclostratigraphy of the Katian Stage (Upper Ordovician) at the Wawenlatu section in Sichuan Province, Southwest China. *Journal of Stratigraphy*, 48: 253–278.
- 56. Pingping Liu, Jiamin Lin, Dianbing Wang, Ben Ma, **Mingsong Li**. **2024**. A tentative study on patterns of volcanic activities and their influencing factors. *Bulletin of Mineralogy, Petrology and Geochemistry*, 43(5): 912–920. <https://doi.org/10.3724/j.issn.1007-2802.20240038>.
- 55. Xu Yao, Shuang Dai, Zhixiang Wang, **Mingsong Li**, Linda Hinnov. **2024**. Astronomical influence on Miocene continental environments in Tibet, east-central Asia. *Geological Society of America Bulletin*, 136(7–8): 3457–3471. <https://doi.org/10.1130/B37129.1>.

2023

54. **Mingsong Li***, Haotian Zhang, Meng Wang, Zhijun Jin. **2023**. Astronomically forced changes in groundwater reservoirs and sea level during the greenhouse world. *Chinese Science Bulletin*, 68(12): 1517–1527. <https://doi.org/10.1360/TB-2022-0790>.
- # 53. **Meng Wang**, **Mingsong Li***, David B. Kemp, Jan Landwehrs, Zhijun Jin. **2023**. Late Triassic sedimentary records reveal the hydrological response to climate forcing and the history of the chaotic Solar System. *Earth and Planetary Science Letters*, 607: 118052. <https://doi.org/10.1016/j.epsl.2023.118052>.
52. Braun Natália Santos, Ciro Climaco Rodrigues, Daniel Ribeiro Franco, **Mingsong Li**, Mariana Aragão Fernandes, Mariane Candido, Raysa de Magalhães Rocha, Thiago Pereira dos Santos, André De Gasperi, Daniel Galvão Carnier Fragoso, Gabriella Fazio, Ana Natalia Gomes Rodrigues. **2023**. Integration of seismic stratigraphy and cyclostratigraphy for high resolution chronostratigraphic correlation: The Albacora Field, Campos Basin, Brazil. *Marine and Petroleum Geology*, 158: 106541. <https://doi.org/10.1016/j.marpetgeo.2023.106541>.
51. Dongyang Liu, Chunju Huang, **Mingsong Li**, Wenhan Chen, Runjian Chu. **2023**. Evolution of phosphorus enrichment in the early to middle Middle Triassic Epoch in Southern China's ancient oceans and its response to astronomical forcing. *Quaternary Sciences*, 43(6): 1538–1546.
- # 50. Meng Wang, Qing Li, Yanguang Dou, Feng Cai, **Mingsong Li**. **2023**. Astronomically forced climate changes in the Okinawa Trough since the Late Pleistocene: Evidence from CSHC-15 core. *Quaternary Sciences*, 43(6): 1711–1721.
49. Rui Zhang, Zhijun Jin, **Mingsong Li**, Quanyou Liu, Peng Li, Ren Wei, Meng Wang, Xiangwu He. **2023**. Astronomical regulation of the Middle Triassic organic-rich shales in the Ordos Basin. *Quaternary Sciences*, 43(6): 1547–1561.
48. Rui Zhang, Zhijun Jin, Rukai Zhu, **Mingsong Li**, Xiao Hui, Ren Wei, Xiangwu He, Qian Zhang. **2023**. Investigation of deposition rate of terrestrial organic-rich shales in China and its implications for shale oil exploration. *Oil & Gas Geology*, 44(4): 829–845.
47. Yiquan Ma, Majie Fan, **Mingsong Li***, James G. Ogg, Chen Zhang, Jun Feng, Chunhua Zhou, Xiaofeng Liu, Yongchao Lu, Huimin Liu, James S. Eldrett, Chao Ma*. **2023**. East Asian lake hydrology modulated by global sea-level variations in the Eocene warmhouse. *Earth and Planetary Science Letters*, 602: 117925. <https://doi.org/10.1016/j.epsl.2022.117925>.
46. Ren Wei, Zhijun Jin, Rui Zhang, **Mingsong Li**, Yongyun Hu, Xiangwu He, Shuai Yuan. **2023**. Orbitally-paced coastal sedimentary records and global sea-level changes in the early Permian. *Earth and Planetary Science Letters*, 620: 118356. <https://doi.org/10.1016/j.epsl.2023.118356>.
45. Rui Zhang, David B. Kemp, Nicolas Thibault, Mads E. Jelby, **Mingsong Li**, Chunju Huang, Yu Sui, Zhixiang Wang, Dongyang Liu, Shizhen Jia. **2023**. Astrochronology and sedimentary noise modeling of Pliensbachian (Early Jurassic) sea-level changes, Paris Basin, France. *Earth and Planetary Science Letters*, 614: 118199. <https://doi.org/10.1016/j.epsl.2023.118199>.
44. Rui Zhang, Zhijun Jin, **Mingsong Li**, Michael Gillman, Shuping Chen, Quanyou Liu, Ren Wei, Juye Shi. **2023**. Long-term periodicity of sedimentary basins in response to astronomical forcing: Review and perspective. *Earth-Science Reviews*, 244: 104533. <https://doi.org/10.1016/j.earscirev.2023.104533>.
43. Chun-Sheng Jin, Deke Xu, **Mingsong Li**, Pengxiang Hu, Zhaoxia Jiang, Jianxing Liu, Yunfa Miao, Fuli Wu, Wentian Liang, Qiang Zhang, Bai Su, Qingsong Liu, Ran Zhang, Jimin Sun. **2023**. Tectonic and orbital forcing of the South Asian monsoon in central Tibet during the late Oligocene. *Proceedings of the National Academy of Sciences*, 120(15): e2214558120. <https://doi.org/10.1073/pnas.2214558120>.
42. Ren Wei, **Mingsong Li**, Rui Zhang, Yongyun Hu, James G. Ogg, Guoyong Liu, He Huang, Xiangwu He, Shuai Yuan, Qifan Lin, Zhijun Jin. **2023**. Obliquity forcing of continental aquifers during the late Paleozoic ice age. *Earth and Planetary Science Letters*, 613: 118174. <https://doi.org/10.1016/j.epsl.2023.118174>.
41. Ren Wei, Rui Zhang, **Mingsong Li**, Xiaojun Wang, Zhijun Jin. **2023**. Obliquity forcing of lake-level changes and organic carbon burial during the Late Paleozoic Ice Age. *Global and Planetary Change*, 223: 104092. <https://doi.org/10.1016/j.gloplacha.2023.104092>.

40. Ruoyuan Qiu, Linhao Fang, Yuanzheng Lu, Yuxuan Chen, Renda Huang, Wenzhi Lei, Pengyuan Zhang, **Mingsong Li**. 2023. Cyclostratigraphy of the Lower Jurassic (Toarcian) terrestrial successions in the Sichuan Basin, southwestern China. *Journal of Asian Earth Sciences*, 250: 105617. <https://doi.org/10.1016/j.jseaes.2023.105617>.

2022

39. **Mingsong Li***, Timothy J. Bralower, Lee R. Kump, Jean M. Self-Trail, James C. Zachos, William D. Rush, Marci M. Robinson. 2022. Astrochronology of the Paleocene-Eocene Thermal Maximum on the Atlantic Coastal Plain. *Nature Communications*, 13(1): 5618. <https://doi.org/10.1038/s41467-022-33390-x>.
38. Tan Zhang, Yifan Li*, Tailiang Fan, Anne-Christine Da Silva, Mingzhi Kuang, Wangwei Liu, Chao Ma, Qi Gao, Juye Shi, Zhiqian Gao, **Mingsong Li***. 2022. Orbital forcing of tropical climate dynamics in the Early Cambrian. *Global and Planetary Change*, 219: 103985. <https://doi.org/10.1016/j.gloplacha.2022.103985>.
37. Jun Tian, Huaichun Wu, Chunju Huang, **Mingsong Li**, Chao Ma, Pinxian Wang. 2022. Revisiting the Milankovitch Theory from the Perspective of the 405 ka Long Eccentricity Cycle. *Earth Science*, 47(10): 3543–3568. <https://doi.org/10.3799/dqkx.2022.248> (Chinese).
36. Jessica E. Tierney, Jiang Zhu, **Mingsong Li**, Andy Ridgwell, Gregory J. Hakim, Christopher J. Poulsen, Ross D. M. Whiteford, James W. B. Rae, Lee R. Kump. 2022. Spatial patterns of climate change across the Paleocene–Eocene Thermal Maximum. *Proceedings of the National Academy of Sciences*, 119(42): e2205326119. <https://doi.org/10.1073/pnas.2205326119>.
35. Zhifeng Zhang, Yongjian Huang, **Mingsong Li**, Xiang Li, Pengcheng Ju, Chengshan Wang. 2022. Obliquity-forced aquifer-eustasy during the Late Cretaceous greenhouse world. *Earth and Planetary Science Letters*, 596: 117800. <https://doi.org/10.1016/j.epsl.2022.117800>.
- # 34. **Yujing Wu**, Xianjun Fang, Linhua Jiang, Biao Song, Baofu Han, **Mingsong Li***, Jianqing Ji*. 2022. Very long-term periodicity of episodic zircon production and Earth system evolution. *Earth-Science Reviews*, 233: 104164. <https://doi.org/10.1016/j.earscirev.2022.104164>.
33. Xu Yao, Shuang Dai, **Mingsong Li**, Linda Hinnov. 2022. Orbital eccentricity and inclination metronomes in Middle Miocene lacustrine mudstones of Jiuxi Basin, Tibet: Closing an astrochronology time gap and calibrating global cooling events. *Global and Planetary Change*, 215: 103896. <https://doi.org/10.1016/j.gloplacha.2022.103896>.
32. Tan Zhang, Yifan Li*, Tailiang Fan, Anne-Christine Da Silva, Juye Shi, Qi Gao, Mingzhi Kuang, Wangwei Liu, Zhiqian Gao, **Mingsong Li***. 2022. Orbitally-paced climate change in the early Cambrian and its implications for the history of the Solar System. *Earth and Planetary Science Letters*, 583: 117420. <https://doi.org/10.1016/j.epsl.2022.117420>.
31. Simin Jin, David B. Kemp, David W. Jolley, Manuel Vieira, James C. Zachos, Chunju Huang, **Mingsong Li**, Wenhan Chen. 2022. Large-scale, astronomically paced sediment input to the North Sea Basin during the Paleocene Eocene Thermal Maximum. *Earth and Planetary Science Letters*, 579: 117340. <https://doi.org/10.1016/j.epsl.2021.117340>.
- # 30. **Meng Wang**, **Mingsong Li***, David B. Kemp, Slah Boulila, James G. Ogg. 2022. Sedimentary noise modeling of lake-level change in the Late Triassic Newark Basin of North America. *Global and Planetary Change*, 208: 103706. <https://doi.org/10.1016/j.gloplacha.2021.103706>.

2021

29. Ying Cui*, **Mingsong Li***, Elsbeth E. van Soelen, Francien Peterse, Wolfram M. Kürschner*. 2021. Massive and rapid predominantly volcanic CO₂ emission during the end-Permian mass extinction. *Proceedings of the National Academy of Sciences*, 118(37): e2014701118. <https://doi.org/10.1073/pnas.2014701118>.

28. Dongyang Liu, Chunju Huang, James G. Ogg, David B. Kemp, **Mingsong Li**, Meiyi Yu, William J. Foster. **2021**. Astronomically forced changes in chemical weathering and redox during the Anisian (Middle Triassic): Implications for marine ecosystem recovery following the end-Permian mass extinction. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 569: 110355. <https://doi.org/10.1016/j.palaeo.2021.110355>.
27. Dongyang Liu, Chunju Huang, David B. Kemp, **Mingsong Li**, James G. Ogg, Meiyi Yu, William J. Foster. **2021**. Paleoclimate and sea level response to orbital forcing in the Middle Triassic of the eastern Tethys. *Global and Planetary Change*, 199: 103454. <https://doi.org/10.1016/j.gloplacha.2021.103454>.
26. Omid Falahatkhah, Ali Kadkhodaie*, Ali A. Ciabeghods, **Mingsong Li***. **2021**. Cyclostratigraphy of the Lower Triassic Kangan Formation in the Salman gas field, eastern Persian Gulf, Iran. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 561: 110045. <https://doi.org/10.1016/j.palaeo.2020.110045>.

2017–2020

25. Haocheng Yu, Kunfeng Qiu, **Mingsong Li**, M. Santosh, Z. G. Zhao, Y. Q. Huang. **2020**. Record of the late Paleozoic ice age from Tarim, China. *Geochemistry, Geophysics, Geosystems*, 21(11): e2020GC009237. <https://doi.org/10.1029/2020GC009237>.
24. Zheng Gong, **Mingsong Li**. **2020**. Astrochronology of the Ediacaran Shuram carbon isotope excursion, Oman. *Earth and Planetary Science Letters*, 547: 116462. <https://doi.org/10.1016/j.epsl.2020.116462>.
23. J. Fred Read, **Mingsong Li**, Linda A. Hinnov, Campbell S. Nelson, Steven Hood. **2020**. Testing for astronomical forcing of cycles and gamma ray signals in outer shelf/upper slope, mixed siliciclastic-carbonates: Upper Oligocene, New Zealand. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 555: 109821. <https://doi.org/10.1016/j.palaeo.2020.109821>.
22. Yang Zhang, James G. Ogg, Matthias Franz, Gerhard H. Bachmann, Michael Szurlies, Heinz-Gerd Röhling, **Mingsong Li**, Christian Rolf, Karsten Obst. **2020**. Carnian (Late Triassic) magnetostratigraphy from the Germanic Basin allowing global correlation of the Mid-Carnian Episode. *Earth and Planetary Science Letters*, 541: 116275. <https://doi.org/10.1016/j.epsl.2020.116275>.
21. Chao Ma, **Mingsong Li**. **2020**. Astronomical time scale of the Turonian constrained by multiple paleoclimate proxies. *Geoscience Frontiers*, 11: 1345–1352. <https://doi.org/10.1016/j.gsf.2020.01.013>.
- # 20. Meng Wang, Honghan Chen*, Chunju Huang, David B. Kemp, Tianwu Xu, Hongan Zhang, **Mingsong Li***. **2020**. Astronomical forcing and sedimentary noise modeling of lake-level changes in the Paleogene Dongpu Depression of North China. *Earth and Planetary Science Letters*, 535: 116116. <https://doi.org/10.1016/j.epsl.2020.116116>.
19. Tan Zhang, Changmin Zhang, Tailiang Fan, Lei Zhang, Rui Zhu, Jinyu Tao, **Mingsong Li**. **2020**. Cyclostratigraphy of Lower Triassic terrestrial successions in the Junggar Basin, northwestern China. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 539: 109493. <https://doi.org/10.1016/j.palaeo.2019.109493>.
18. **Mingsong Li**, Hubert L. Barnes. **2019**. Orbitally forced sphalerite growth in the Upper Mississippi Valley District. *Geochemical Perspectives Letters*, 12: 18–22. <https://doi.org/10.7185/geochemlet.1929> [Cover story].
17. Peng Gao, Junsheng Nie, **Mingsong Li**, Pu Li. **2019**. Confirmation of a late Miocene subchron C4n.2n-1r from the eastern Qaidam Basin in the NE Tibetan Plateau. *Journal of Geophysical Research: Solid Earth*, 124(12): 12354–12365. <https://doi.org/10.1029/2019JB017936>.
16. Matthias Sinnesael, David De Vleeschouwer, Christian Zeeden, Sietske J. Batenburg, Anne-Christine Da Silva, Niels J. de Winter, Jaume Dinarès-Turell, Anna Joy Drury, Gabriele Gambacorta,

- Frederik J. Hilgen, Linda A. Hinnov, Alexander J. L. Hudson, David B. Kemp, Margriet L. Lantink, Jirí Laurin, **Mingsong Li**, Diederik Liebrand, Chao Ma, Stephen R. Meyers, Johannes Monkenbusch, Alessandro Montanari, Theresa Nohl, Heiko Pälike, Damien Pas, Micha Ruhl, Nicolas Thibault, Maximilian Vahlenkamp, Luis Valero, Sébastien Wouters, Huaichun Wu, Philippe Claeys. **2019**. The Cyclostratigraphy Intercomparison Project (CIP): consistency, merits and pitfalls. *Earth-Science Reviews*, 199: 102965. <https://doi.org/10.1016/j.earscirev.2019.102965>.
15. Qingda Su, Junsheng Nie, Zeng Luo, **Mingsong Li**, Richard Heermance, Carmala Garziona. **2019**. Detection of strong precession cycles from the late Pliocene sedimentary records of northeastern Tibetan Plateau. *Geochemistry, Geophysics, Geosystems*, 20(8): 3901–3912. <https://doi.org/10.1029/2019GC008447>.
 14. **Mingsong Li**, Chunju Huang, James Ogg, Yang Zhang, Linda Hinnov, Huaichun Wu, Zhong-Qiang Chen, Zhuoyan Zou. **2019**. Paleoclimate proxies for cyclostratigraphy: Comparative analysis using a Lower Triassic marine section in South China. *Earth-Science Reviews*, 189: 125–146. <https://doi.org/10.1016/j.earscirev.2019.01.011>.
 13. **Mingsong Li**, Linda Hinnov, Lee Kump. **2019**. *Acycle*: Time-series analysis software for paleoclimate research and education. *Computers & Geosciences*, 127: 12–22. <https://doi.org/10.1016/j.cageo.2019.02.011>.
 12. **Mingsong Li**, Linda A. Hinnov, Chunju Huang, James G. Ogg. **2018**. Sedimentary noise and sea levels linked to land–ocean water exchange and obliquity forcing. *Nature Communications*, 9: 1004. <https://doi.org/10.1038/s41467-018-03454-y>.
 11. **Mingsong Li**, Lee R. Kump, Linda A. Hinnov, Michael E. Mann. **2018**. Tracking variable sedimentation rates and astronomical forcing in Phanerozoic paleoclimate proxy series with evolutionary correlation coefficients and hypothesis testing. *Earth and Planetary Science Letters*, 501: 165–179. <https://doi.org/10.1016/j.epsl.2018.08.041>.
 10. **Mingsong Li**, Chunju Huang, Linda Hinnov, Weizhe Chen, James Ogg, Wei Tian. **2018**. Astrochronology of the Anisian stage (Middle Triassic) at the Guandao reference section, South China. *Earth and Planetary Science Letters*, 482: 591–606. <https://doi.org/10.1016/j.epsl.2017.11.042>.
 9. **Mingsong Li**, Yang Zhang, Chunju Huang, James Ogg, Linda Hinnov, Yongdong Wang, Zhuoyan Zou, Liqin Li. **2017**. Astronomical tuning and magnetostratigraphy of the Upper Triassic Xujiahe Formation of South China and Newark Supergroup of North America: Implications for the Late Triassic time scale. *Earth and Planetary Science Letters*, 475: 207–223. <https://doi.org/10.1016/j.epsl.2017.07.015>.

2010–2016

8. **Mingsong Li**, Chunju Huang, Linda Hinnov, James Ogg, Zhong-Qiang Chen, Yang Zhang. **2016**. Obliquity-forced climate during the Early Triassic hothouse in China. *Geology*, 44(8): 623–626. <https://doi.org/10.1130/G37970.1> [*Cover story*].
7. **Mingsong Li**, James Ogg, Yang Zhang, Chunju Huang, Linda Hinnov, Zhong-Qiang Chen, Zhuoyan Zou. **2016**. Astronomical tuning of the end-Permian extinction and the Early Triassic Epoch of South China and Germany. *Earth and Planetary Science Letters*, 441: 10–25. <https://doi.org/10.1016/j.epsl.2016.02.017>.
6. Zhuoyan Zou, Chunju Huang, **Mingsong Li**, Yang Zhang. **2016**. Climate change response to astronomical forcing during the Oligocene-Miocene transition in the equatorial Atlantic (ODP Site 926). *Science China Earth Sciences*, 59(8): 1665–1673. <https://doi.org/10.1007/s11430-016-5311-y>.
5. Yang Zhang, **Mingsong Li**, James G. Ogg, Paul Montgomery, Chunju Huang, Zhong-Qiang Chen, Zhiqiang Shi, Paul Enos, Daniel J. Lehrmann. **2015**. Cycle-calibrated magnetostratigraphy of middle Carnian from South China: Implications for Late Triassic time scale and termination of the Yangtze Platform. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 436: 135–166. <https://doi.org/10.1016/j.palaeo.2015.05.033>.

4. Yuewu Sun, **Mingsong Li**, Wenchun Ge, Yanlong Zhang, Dejun Zhang. **2013**. Eastward termination of the Solonker–Xar Moron River Suture determined by detrital zircon U–Pb isotopic dating and Permian floristics. *Journal of Asian Earth Sciences*, 75: 243–250. <https://doi.org/10.1016/j.jseaes.2013.07.018>.
3. Yuewu Sun, **Mingsong Li**, Guowei Zhao. **2012**. A new terrestrial lithostratigraphic unit of the Cisuralian (Early Permian) in the Yanbian area, Jilin Province. *Journal of Stratigraphy*, 36: 89–96.
2. **Mingsong Li**, Yuewu Sun, Guowei Zhao. **2011**. Discovery of Early Permian Cathaysia flora from Daxinggou of Wangqing County, Yanbian area, Jilin Province, China and its geological significance. *Advances in Earth Science*, 26: 339–346.
1. Shuqin Zhang, Yuying Liu, **Mingsong Li**. **2010**. Eocene-Miocene palynological assemblages in Wanchang area of Jilin and their stratigraphic significance. *Global Geology*, 29: 357–362.

5. Patents and Software Copyrights

3. **Mingsong Li**. Time Series Analysis Software V2.8. Computer Software Copyright, Registration No. 2026SR0541650, **2026**.
2. Meng Wang, **Mingsong Li**. AcyclePy Time Series and Earth Science Data Analysis Software V1.0. Computer Software Copyright, Registration No. 2026SR0970480, **2026**.
- # 1. Jiahao Liu, **Mingsong Li***, Xinwen Zhang. A Paleoclimate Reconstruction Method Based on Isotope Data Assimilation and Related Devices [Invention Patent Application]. Application No. 202511982369.1; Document Issue No. 2026033100373820. Applicant: Peking University, **2025**.

6. Software Packages

Acycle: Time-series analysis software for paleoclimate research and education

<https://github.com/mingsongli/acycle> or <https://acycle.org>

A comprehensive and easy-to-use desktop application for macOS and Windows

Supporting the research of more than 1,000 scientists at institutions including the University of Cambridge, Yale University, Princeton University, Columbia University, the University of Calgary, the Australian National University, the Chinese Academy of Sciences, Utrecht University, the University of Padova, the University of Zaragoza, and the National Observatory (Brazil).

DeepDA: Paleoclimate data assimilation software

<http://github.com/mingsongli/deepda>

7. Conference Abstracts

2025

136. **Mingsong Li**. 2025. Data assimilation of carbon-cycle perturbations during the Paleocene–Eocene Thermal Maximum. The 3rd Forum on Paleoclimate Modeling and Symposium on Discipline Development. Xi'an, Shaanxi, China. [Oral]
135. **Mingsong Li**. 2025. Artificial intelligence reshaping the geological concept of space and time. The 441st Shuangqing Forum of the National Natural Science Foundation of China. Beijing, China. [Invited Oral]
- # 134. Jiahao Liu, **Mingsong Li**. 2025. Carbon isotope data assimilation for the Paleocene–Eocene Thermal Maximum. The 3rd National Symposium on Data-Driven Development in Geosciences. Zhuhai, Guangdong, China. [Oral]
133. Ren Wei, Zhijun Jin, **Mingsong Li**, Yongyun Hu, Rui Zhang. 2025. Orbital-scale variability in atmospheric meridional heat transport during the Late Paleozoic Ice Age. The 8th Conference on Earth System Science. Shanghai, China. [Oral]

- # 132. Qingqing Jiang, **Mingsong Li**. 2025. Climate–carbon cycle responses to orbital climate variability in the Early Eocene based on the Earth system model cGENIE. The 8th Conference on Earth System Science. Shanghai, China. [Oral]
- 131. Jian Chen, **Mingsong Li**, Yige Zhang. 2025. Modulation of Late Devonian sea-level fluctuations by the 1-Myr obliquity modulation cycle. The 8th Conference on Earth System Science. Shanghai, China. [Oral]
- 130. Renda Huang, **Mingsong Li**, Ruoyuan Qiu, Youwei Wang, Rui Zhang, Zhijun Jin. 2025. Enhancing astronomical cycle detection through multi-series stacking and averaging. The 8th Conference on Earth System Science. Shanghai, China. [Oral]
- # 129. Hanyu Zhu, **Mingsong Li**, Junpeng Zhang, Kaixuan Ji, Meng Wang. 2025. Astrochronologic constraints on the duration of the Hirnantian Stage. The 8th Conference on Earth System Science. Shanghai, China. [Oral]
- 128. Meng Wang, **Mingsong Li**, Elizabeth A. Hajek, David B. Kemp, Zhijun Jin. 2025. Sedimentary preservation of orbital signals and the reliability of astronomical time scales. The 8th Conference on Earth System Science. Shanghai, China. [Oral]
- 127. Meng Wang, **Mingsong Li**. 2025. Sedimentary records and paleoclimate modeling evidence for Late Triassic lake-level evolution. The 8th Conference on Earth System Science. Shanghai, China. [Oral]
- # 126. Hanyu Zhu, **Mingsong Li**. 2025. A new method for sedimentation-rate detection in high-resolution cyclostratigraphic data based on cross-spectral estimation. The 8th National Congress on Sedimentology. Beijing, China. [Oral]
- # 125. Jiahao Liu, **Mingsong Li**. 2025. Carbon isotope data assimilation for the Paleocene–Eocene Thermal Maximum. The 8th National Congress on Sedimentology. Beijing, China. [Oral]
- # 124. Xiaoyu Zhang, Kaixuan Ji, **Mingsong Li**. 2025. Astronomical forcing of mid-Proterozoic Earth system evolution: a case study of the Chuanlinggou Formation, North China. The 8th National Congress on Sedimentology. Beijing, China. [Oral]
- # 123. Zhixin Wang, **Mingsong Li**. 2025. Sedimentary rhythms of the Martian Noachian based on statistical tuning techniques. The 8th National Congress on Sedimentology. Beijing, China. [Oral]
- # 122. Qingqing Jiang, **Mingsong Li**. 2025. Orbital forcing and climate responses during the PETM hyperthermal: an integrated study based on Earth system modeling and cyclostratigraphy at ODP Site 1172. The 8th National Congress on Sedimentology. Beijing, China. [Oral]
- # 121. Kaixuan Ji, Paul B. Wignall, Jinnan Tong, Yongyun Hu, **Mingsong Li**, Daoliang Chu. 2025. Sea-level control on Late Permian–Early Triassic continental strata in North China and global stratigraphic correlation. The 8th National Congress on Sedimentology. Beijing, China. [Oral]
- # 120. Kaixuan Ji, **Mingsong Li**, Yongyun Hu, Haotian Zhang, Fanhao Gong, Shuai Yuan. 2025. Late Cretaceous paleowater-depth changes regulated by the 1.2-Myr obliquity modulation cycle. The 8th National Congress on Sedimentology. Beijing, China. [Oral]
- # 119. Haotian Zhang, Kaixuan Ji, Yan Chen, James Ogg, Zhiming Sun, Paul Wignall, Meng Wang, Haoxun Zhang, Xiaoyu Zhang, Yang Zhang, Keke Huang, **Mingsong Li**. 2025. Astronomically calibrated integrated stratigraphy of the Induan Stage in the aftermath of the Permian–Triassic mass extinction. EGU General Assembly. Vienna, Austria. [Oral]
- 118. Qian He, Sofie Lindström, Stephen Hesselbo, Christian Bjerrum, **Mingsong Li**, Jianxin Yu, Jun Shen. 2025. Hydrological cycle perturbations of continental weathering during the Triassic–Jurassic transition and the Toarcian Oceanic Anoxic Event. EGU General Assembly. Vienna, Austria. [Oral]
- 117. **Mingsong Li**. 2025. Carbon cycle data assimilation during the Paleocene–Eocene Thermal Maximum. International Symposium on the Tethys Evolution and Biotic and Environmental Responses. Beijing, China. [Invited Plenary Lecture]

116. **Mingsong Li**. 2025. Astrochronology and global sea-level change. Geochronology Gordon Research Conference. Newry, Maine, USA. [Invited Oral]
115. **Mingsong Li**. 2025. Cyclostratigraphic advances in the study of ancient hydrological change. Forum on Deep-Water Sedimentation and Deep Oil and Gas Exploration. Qingdao, Shandong, China. [Invited Oral]
114. **Mingsong Li**. 2025. Triassic astro-magnetostratigraphy and paleoclimate change. 2025 Annual Meeting of the Beijing Geophysical Society. Beijing, China. [Invited Oral]

2024

113. **Mingsong Li**, Jiahao Liu, Haoxun Zhang, Xinwen Zhang, Lee R. Kump, Andy Ridgwell. 2024. Carbon cycle data assimilation during the Paleocene–Eocene Thermal Maximum and the Early Eocene Climatic Optimum. AGU Fall Meeting. Washington, DC, USA. [Invited Oral]
- # 112. Qingqing Jiang, **Mingsong Li**. 2024. Astrochronology of the Paleocene–Eocene Thermal Maximum at ODP Site 1172. AGU Fall Meeting. Washington, DC, USA.
- # 111. Kaixuan Ji, **Mingsong Li**. 2024. Applications of sedimentology and datafication in astrochronology. AGU Fall Meeting. Washington, DC, USA.
110. Meng Wang, **Mingsong Li**, Elizabeth Hajek, David B. Kemp, Yujing Wu, Hanyu Zhu, Zhijun Jin. 2024. Preservation of orbital signals in the sedimentary record and the robustness of astronomical time scales. AGU Fall Meeting. Washington, DC, USA.
109. Jian Chen, **Mingsong Li**, Kunyuan Ma, Yuehan Lu. 2024. Cyclostratigraphic calibration of the Late Devonian and its implications for biocrises and bioevents in the Illinois Basin, eastern United States. GSA Annual Meeting. Seattle, USA.
108. **Mingsong Li**, Haoxun Zhang, Yongyun Hu, Jiahao Liu, Hanyu Zhu, Xinwen Zhang. 2024. Paleoclimate data assimilation: principles, progress, and prospects. The 2nd National Symposium on Data-Driven Development in Geosciences. Beijing, China. [Invited Oral]
- # 107. Hanyu Zhu, Junpeng Zhang, **Mingsong Li**. 2024. Preliminary optimization of high-throughput analysis for large cyclostratigraphic datasets: examples from high-resolution XRF-CS series and sediment-model-generated series. The 2nd National Symposium on Data-Driven Development in Geosciences. Beijing, China.
106. Meng Wang, Elizabeth A. Hajek, David B. Kemp, Yujing Wu, Hanyu Zhu, Zhijun Jin, **Mingsong Li**. 2024. Big-data analysis of stochastic sedimentation models reveals sedimentary preservation of orbital signals. The 2nd National Symposium on Data-Driven Development in Geosciences. Beijing, China.
- # 105. Haoxun Zhang, **Mingsong Li**. 2024. Reconstructing the greenhouse climate state of the Early Eocene Climatic Optimum using paleoclimate data assimilation. The 9th Young Geoscientists Forum. Xiamen, China.
- # 104. Zhixin Wang, **Mingsong Li**. 2024. Sedimentary rhythms during the Martian Noachian. The 9th Young Geoscientists Forum. Xiamen, China.
- # 103. Hanyu Zhu, **Mingsong Li**, Junpeng Zhang. 2024. A new floating astronomical time scale of the Hirnantian Stage in the Upper Yangtze Region of South China. The 6th International Conference on Palaeogeography. Nanjing, China.
102. Rui Zhang, David B. Kemp, Nicolas Thibault, Mads E. Jelby, **Mingsong Li**, Chunju Huang. 2024. Astrochronology and sedimentary noise modeling of Pliensbachian (Early Jurassic) sea-level changes, Paris Basin, France. The 6th International Conference on Palaeogeography. Nanjing, China.
- # 101. Xiaoyu Zhang, Kaixuan Ji, **Mingsong Li**. 2024. Astronomical forcing of the mid-Proterozoic climate: evidence from the Chuanlinggou Formation in North China. The 6th International Conference on Palaeogeography. Nanjing, China.

- # 100. Meng Wang, **Mingsong Li**, David B. Kemp, Jan Landwehrs, Zhijun Jin. 2024. Sedimentary records and climate simulation evidence of Late Triassic hydrological change. The 6th International Conference on Palaeogeography. Nanjing, China.
- # 99. Qingqing Jiang, **Mingsong Li**. 2024. Astrochronology of the Paleocene–Eocene Thermal Maximum at ODP Site 1172. The 6th International Conference on Palaeogeography. Nanjing, China.
- # 98. Kaixuan Ji, **Mingsong Li**, Fanhao Gong, Haotian Zhang, Shuai Yuan, Dejun Zhang. 2024. High-resolution sea- and lake-level reconstructions of the Late Cretaceous imply a “Zero-Sum Dynamics” ocean–land water circulation model. EGU General Assembly. Vienna, Austria.
- # 97. Meng Wang, **Mingsong Li**, Elizabeth A. Hajek, David B. Kemp, Yujing Wu, Hanyu Zhu, Zhijun Jin. 2024. Detection of orbital signals in the sedimentary record through stochastic statistical modeling. EGU General Assembly. Vienna, Austria.

2023

- 96. Xu Yao, Shuang Dai, **Mingsong Li**, Linda A. Hinnov. 2023. Did astronomical forcing trigger abrupt Cenozoic global cooling events? AGU Fall Meeting. San Francisco, CA, USA & Online. [Poster]
- # 95. Meng Wang, **Mingsong Li**, Elizabeth A. Hajek, David B. Kemp, Yujing Wu, Zhijun Jin, Hanyu Zhu. 2023. Detecting astronomical signals in the sedimentary record and the robustness of astronomical time scales. AGU Fall Meeting. San Francisco, CA, USA & Online. [Poster]
- 94. Ren Wei, Zhijun Jin, **Mingsong Li**. 2023. Orbitally paced continental aquifers and global sea-level changes during the Late Paleozoic Ice Age. AGU Fall Meeting. San Francisco, CA, USA & Online. [Poster]
- 93. **Mingsong Li**, Lee R. Kump, Andy Ridgwell, Jessica E. Tierney, Gregory J. Hakim, Steven B. Malevich, Christopher J. Poulsen, Robert Tardif, Jiang Zhu. 2023. Reconstruction of carbon cycle perturbations during the Paleocene–Eocene Thermal Maximum using data assimilation. AGU Fall Meeting. San Francisco, CA, USA & Online. [Oral]
- # 92. Qingqing Jiang, **Mingsong Li**. 2023. Astrochronology of the Paleocene–Eocene Thermal Maximum in ODP Hole 1172D. The 31st Annual Meeting of the Palaeontological Society of China. Nanjing, China. [Oral]
- # 91. Kaixuan Ji, **Mingsong Li**, Fanhao Gong, Dejun Zhang. 2023. Astronomical forcing and paleoenvironmental responses of the Upper Cretaceous Mingshui Formation in the Songliao Basin. The 31st Annual Meeting of the Palaeontological Society of China. Nanjing, China. [Oral]
- # 90. Haotian Zhang, Kaixuan Ji, Yan Chen, James G. Ogg, Haoxun Zhang, Xiaoyu Zhang, Zhiming Sun, Hanyu Zhu, Meng Wang, Yang Zhang, Keke Huang, **Mingsong Li**. 2023. Early Triassic climate change driven by the 1.2-Myr obliquity cycle. The 31st Annual Meeting of the Palaeontological Society of China. Nanjing, China. [Oral]
- # 89. Hanyu Zhu, **Mingsong Li**, Kaixuan Ji, Junpeng Zhang. 2023. A floating astronomical time scale for the Hirnantian Stage in the Upper Yangtze Region of South China. The 31st Annual Meeting of the Palaeontological Society of China. Nanjing, China. [Oral]
- 88. **Mingsong Li**. 2023. Applications of Acycle 2.7 in deep-time environmental change research and education. The 31st Annual Meeting of the Palaeontological Society of China. Nanjing, China. [Keynote]
- 87. **Mingsong Li**, Chunju Huang, Linda Hinnov, Meng Wang, Lee Kump, Michael Mann, Rui Zhang, Zhijun Jin. 2023. The obscured 405-kyr eccentricity cycle and the challenge of standardizing astrochronologic studies. Symposium on Earth Rhythms, Astronomical Cycles, and Zhang Yiwei’s Academic Thought on Crustal Oscillation. Beijing, China. [Oral]
- 86. **Mingsong Li**. 2023. Reconstructing the spatiotemporal evolution of the Eocene carbon cycle using astrochronology and data assimilation. The 7th Conference on Earth Systems Science – Young Scholars Forum. Shanghai, China. [Invited Oral]

85. **Mingsong Li**, Haotian Zhang, Meng Wang, Ren Wei, Rui Zhang, Zhijun Jin. 2023. Astronomical mechanisms linking groundwater reservoirs and long-term sea-level change during greenhouse climates. The 7th Conference on Earth Systems Science. Shanghai, China. [Oral]
- # 84. Meng Wang, **Mingsong Li**, David B. Kemp, Jan Landwehrs, Zhijun Jin. 2023. Sedimentary records and paleoclimate simulation evidence of Late Triassic paleohydrological evolution. The 7th Conference on Earth Systems Science. Shanghai, China. [Oral]
- # 83. Haotian Zhang, **Mingsong Li**, Haoxun Zhang, Xiaoyu Zhang, Kaixuan Ji, Hanyu Zhu, Meng Wang. 2023. Early Triassic climate change driven by the 1.2-Myr obliquity cycle. The 7th Conference on Earth Systems Science. Shanghai, China. [Oral]
- # 82. Haoxun Zhang, **Mingsong Li**. 2023. Reconstructing the greenhouse climate state of the Early Eocene Climatic Optimum using paleoclimate data assimilation. The 7th Conference on Earth Systems Science. Shanghai, China. [Oral]
- # 81. Xiaoyu Zhang, **Mingsong Li**. 2023. Progress and prospects in the study of Mesoproterozoic astronomical cycles. Symposium on Mesoproterozoic Geology and Environment of the Southern North China Craton. Zhengzhou, China. [Oral]
80. Zhijun Jin, Shuping Chen, Rui Zhang, **Mingsong Li**, Juye Shi, Ren Wei, Guochen Liu, Jingchang Li, Xiaopeng Gao, Baonian Wu, Xiangwu He. 2023. Responses of sedimentary basin fluctuation processes to astronomical cycles. The 7th National Congress on Sedimentology. Chengdu, China. [Oral]
- # 79. Haoxun Zhang, **Mingsong Li**. 2023. Reconstructing Early Eocene Climatic Optimum sea-surface temperatures using paleoclimate data assimilation. The 7th National Congress on Sedimentology. Chengdu, China. [Oral]
- # 78. Haotian Zhang, **Mingsong Li**. 2023. High-resolution reconstruction of lake-level changes in continental basins and their astronomical forcing. The 7th National Congress on Sedimentology. Chengdu, China. [Oral]
- # 77. Meng Wang, **Mingsong Li**, David B. Kemp, Slah Boulila, James G. Ogg. 2023. High-resolution lake-level changes in continental basins and their astronomical forcing. The 7th National Congress on Sedimentology. Chengdu, China. [Oral]
76. **Mingsong Li**, Timothy J. Bralower, Lee R. Kump, Jean M. Self-Trail, James C. Zachos, William D. Rush, Marci M. Robinson. 2023. Astrochronology and paleoenvironmental change of the Paleocene–Eocene Thermal Maximum on the Atlantic Coastal Plain. The 7th National Congress on Sedimentology. Chengdu, China. [Oral]
75. **Mingsong Li**, Timothy J. Bralower, Lee R. Kump, Jean M. Self-Trail, James C. Zachos, William D. Rush, Marci M. Robinson. 2023. Astronomical calibration of the Paleocene–Eocene Thermal Maximum on the Atlantic Coastal Plain. EGU General Assembly. Vienna, Austria. [Invited Oral]

2022

74. Ciro Rodrigues, Natália Braun dos Santos, Daniel Ribeiro Franco, Mariana Aragão Fernandes, **Mingsong Li**, Raysa Magalhães Rocha, Mariane Candido, Daniel Galvão Fragoso, André De Gasperi. 2022. Cyclostratigraphy of sedimentary records from Albacora oilfield (Campos Basin, Southern Brazil). AGU Fall Meeting, Chicago & Online Everywhere.
73. **Mingsong Li**, Timothy J. Bralower, Lee R. Kump, Jean M. Self-Trail, James C. Zachos, William D. Rush, Marci M. Robinson. 2022. Cyclostratigraphy of the Paleocene–Eocene Thermal Maximum on the Atlantic Coastal Plain: a data–model comparison. AGU Fall Meeting, Chicago & Online Everywhere. [Poster]
72. Xu Yao, Shuang Dai, **Mingsong Li**, Linda Hinnov. 2022. Astronomical forcing of Miocene red and green mudstone rhythms reveals a negative hydrologic budget linked to orbital eccentricity maxima in northeastern Tibet. AGU Fall Meeting, Chicago & Online Everywhere. [Poster]

- # 71. **Meng Wang, Mingsong Li**, David B. Kemp, Slah Boulila, James G. Ogg. 2022. Sedimentary noise modeling of lake-level change and astronomical forcing in the Late Triassic Newark Basin of North America. The 21st International Sedimentological Congress, Beijing. [Oral]
- # 70. **Haotian Zhang, Mingsong Li**. 2022. Astrochronology of the Induan of the Lower Triassic: evidence from the Germanic Basin. AGU Fall Meeting, Chicago & Online Everywhere. [Poster]
- # 69. **Meng Wang, Mingsong Li**, David B. Kemp. 2022. Astronomical recalibration of the Upper Triassic in St. Audrie's Bay and its implication for chaotic behavior in the solar system. AGU Fall Meeting, Chicago & Online Everywhere. [Poster]
- 68. **Mingsong Li**, Lee Robert Kump, Andy Ridgwell, Jessica E. Tierney, Gregory J. Hakim, Steven B. Malevich, Christopher J. Poulsen, Robert Tardif, Jiang Zhu. 2022. Global warming and ocean acidification during the Paleocene–Eocene Thermal Maximum. 2022 GENIE/muffin symposium. Online. [Oral]

2021

- 67. Jessica E. Tierney, Jiang Zhu, **Mingsong Li**, Andy Ridgwell, Lee Robert Kump, Gregory J. Hakim, Christopher J. Poulsen. 2021. Spatial patterns of temperature and hydroclimate during the Paleocene–Eocene Thermal Maximum: Fingerprints of a warmer world. AGU Fall Meeting, New Orleans, LA & Online Everywhere.
- 66. Yiquan Ma, Majie Fan, James Ogg, **Mingsong Li**, Chao Ma. 2021. Orbitally tuned terrestrial time scale and astronomical forcing of regional lake hydrology change in East Asia during the middle–late Eocene. AGU Fall Meeting, New Orleans, LA & Online Everywhere.
- 65. Xu Yao, Shuang Dai, **Mingsong Li**, Linda Hinnov. 2021. Cyclostratigraphy of early–middle Miocene lacustrine successions in Jiuxi Basin, northeastern Tibet. AGU Fall Meeting, New Orleans, LA & Online Everywhere.
- 64. **Mingsong Li**, Lee Robert Kump, Andy Ridgwell, Jessica E. Tierney, Gregory J. Hakim, Steven B. Malevich, Christopher J. Poulsen, Robert Tardif, Jiang Zhu. 2021. Global warming and ocean acidification during the Paleocene–Eocene Thermal Maximum. AGU Fall Meeting, New Orleans, LA & Online Everywhere.
- 63. **Mingsong Li**, Meng Wang, James Ogg, Chunju Huang, Linda Hinnov. 2021. Astronomically driven paleolake- and sea-level changes in the Triassic. The 3rd Young Scientist Forum of Earth Science. Guiyang, Guizhou.
- # 62. Haotian Zhang, **Mingsong Li**. 2021. Astrochronology of the Induan Stage (Lower Triassic) in the Germanic Basin based on the 405-kyr tuning. The 6th Conference on Earth System Science. Shanghai.
- 61. **Mingsong Li**, Chunju Huang, Linda Hinnov, Meng Wang, Lee Kump, Michael Mann. 2021. The obscuring of the 405-kyr long eccentricity cycle and the dilemma of cyclostratigraphy. The 6th Conference on Earth System Science. Shanghai.
- # 60. **Meng Wang, Mingsong Li**, David B. Kemp, Slah Boulila. 2021. Glacially driven sea-level changes in the Late Triassic. vEGU General Assembly. EGU21-6117.

2020

- 59. Peng Gao, Junsheng Nie, **Mingsong Li**, Pu Li. 2020. Confirmation of a Late Miocene Subchron C4n.2n-1r from the Eastern Qaidam Basin in the NE Tibetan Plateau. Geological Society of America Annual Meeting.
- # 58. **Meng Wang, Mingsong Li**. 2020. Reconstruction of high-resolution lake level and its astronomical forcing during the Paleogene. GeoUtrecht Virtual 2020.
- 57. **Mingsong Li**, Hubert L. Barnes. 2020. Astronomically forced sphalerite growth and groundwater circulation in the Upper Mississippi Valley District, USA. GeoUtrecht Virtual 2020.

56. **Mingsong Li**, Hubert L. Barnes. 2020. Orbitally forced groundwater circulation and sphalerite growth in the Upper Mississippi Valley District. Goldschmidt Virtual 2020.
55. **Mingsong Li**, Lee R. Kump, Oluwaseyi Ajayi, Daniel Amrhein, Gregory J. Hakim, Steven B. Malevich, Chris Poulsen, Andy Ridgwell, Robert Tardif, Jessica E. Tierney, Fuqing Zhang, Jiang Zhu. 2020. Impact of new observations on improved understanding of the Paleocene–Eocene Thermal Maximum. Goldschmidt Virtual 2020.
54. Thomas Wonik, Arne Ulfers, Matthias Sinnesael, **Mingsong Li**, Christian Zeeden. 2020. Detecting and using Milankovic cycles in borehole logging data: Comparing methods and application to Lake Ohrid. EGU General Assembly. Vienna, Austria.
53. Christian Zeeden, Matthias Sinnesael, **Mingsong Li**, Arne Ulfers, Thomas Wonik. 2020. Systematically probing for Milanković cycles in borehole logging data. The 36th International Geological Congress, Delhi, India.

2019

52. Christian Zeeden, Matthias Sinnesael, **Mingsong Li**, Arne Ulfers, Thomas Wonik. 2019. Borehole log cyclostratigraphy: towards systematic probing for Milankovic cycles in logging data. *Geophysical Research Abstracts*, 21.
51. J. Fred Read, **Mingsong Li**, Linda A. Hinnov, Campbell S. Nelson, Steven D. Hood. 2019. Are Milankovitch cycles evident in the gamma ray logs of Late Oligocene mixed siliciclastic-carbonates of Tikorangi Formation, Taranaki Peninsula, New Zealand? In: P. J. J. Kamp and A. Pittari (eds.). Abstract Volume: Geosciences 2019, Hamilton, New Zealand. Geoscience Society of New Zealand Miscellaneous Publication 154A. p. 144.
50. **Mingsong Li**, Lee R. Kump, Fuqing Zhang, Andy Ridgwell, Gregory J. Hakim, Jessica E. Tierney. 2019. Assessing the impact of new observations on improved understanding of the Paleocene–Eocene Thermal Maximum. AGU Fall Meeting, San Francisco, USA.
- # 49. **Meng Wang**, Honghan Chen, **Mingsong Li**. 2019. Tracking Paleogene lake-level change and its long-term cyclicities by using sedimentary noise. AGU Fall Meeting, San Francisco, USA.
48. Daniel Amrhein, Gregory Hakim, Lee Kump, **Mingsong Li**, Steven Malevich, Chris Poulsen, Andy Ridgwell, Robert Tardif, Jessica Tierney, Jiang Zhu. 2019. Tropical Variability from the Last Glacial Maximum to Present. AGU Fall Meeting, San Francisco, USA.
47. Jessica Tierney, Steven Malevich, Jiang Zhu, Jonathan King, Chris Poulsen, Andy Ridgwell, **Mingsong Li**, Lee Kump, Robert Tardif, Greg Hakim. 2019. Glacial cooling and climate sensitivity revisited. AGU Fall Meeting, San Francisco, USA.
46. Jessica Tierney, Jiang Zhu, Jonathan King, **Mingsong Li**, Steven Malevich, Chris Poulsen, Andy Ridgwell, Greg Hakim, Robert Tardif, Lee Kump. 2019. A new view of the Eocene greenhouse world from paleoclimate data assimilation. AGU Fall Meeting, San Francisco, USA.
45. Christian Zeeden, Matthias Sinnesael, **Mingsong Li**, Arne Ulfers, Thomas Wonik. 2019. Towards systematic probing for Milankovic cycles in borehole logging data and complex settings. AGU Fall Meeting, San Francisco, USA.
44. Hwei Duan, Jiyong Jiang, **Mingsong Li**. 2019. Bridging Chinese Landscape Painting and geosciences. AGU Fall Meeting, San Francisco, USA.
43. Dongyang Liu, Chunju Huang, David Kemp, **Mingsong Li**, Meiyi Yu, William Foster. 2019. Orbital forcing in the Middle Triassic of the eastern Tethys: implications for paleoclimate and eustasy. AGU Fall Meeting, San Francisco, USA.
42. Tan Zhang, Changmin Zhang, **Mingsong Li**, Tailiang Fan. 2019. Cyclostratigraphy of the Lower Triassic terrestrial successions in the Junggar Basin, northwestern China. AGU Fall Meeting, San Francisco, USA.

41. Naihua Xue, Wei Wang, **Mingsong Li**. 2019. Astrochronology and Geochemistry for the Doushantuo Formation of Ediacaran of Yichang in South China. AGU Fall Meeting, San Francisco, USA.
40. Yan Chen, Yang Zhang, James G. Ogg, Haishui Jiang, Zhiming Sun, **Mingsong Li**. 2019. Integrated Early Triassic time scale of geomagnetic polarity, cyclostratigraphy, biostratigraphy and geochemical excursions from South China and its global correlation. AGU Fall Meeting, San Francisco, USA.
39. Ying Cui, Els van Soelen, **Mingsong Li**, Wolfram Kürschner. 2019. Permian–Triassic global marine and terrestrial carbon cycle records from the Finnmark platform in Norway. Geological Society of America Annual Meeting, Phoenix, USA.
38. Jessica E. Tierney, Jonathan King, Steven Brewster Malevich, Tripti Bhattacharya, Ran Feng, Jiang Zhu, **Mingsong Li**, Chris J. Poulsen, Alan M. Haywood, Bette Otto-Bliesner, Andy Ridgwell, Greg Hakim, Robert Tardif, Lee R. Kump. 2019. New Views of Warm Worlds from Paleoclimate Data Assimilation. Goldschmidt 2019, Barcelona, Spain.
37. Christian Zeeden, Matthias Sinnesael, **Mingsong Li**, Arne Ulfers, Thomas Wonik. 2019. Towards systematic probing for Milankovic cycles in borehole logging data and complex settings. STRATI, 3rd International Congress on Stratigraphy. Milano, Italy.
36. Christian Zeeden, Matthias Sinnesael, **Mingsong Li**, Arne Ulfers, Thomas Wonik. 2019. EGU2019-3284: Borehole log cyclostratigraphy: towards systematic probing for Milankovic cycles in logging data. EGU General Assembly. Vienna, Austria.

2018

35. **Mingsong Li**, Chunju Huang, James Ogg, Linda Hinnov, Yang Zhang, Weizhe Chen, Wei Tian. 2018. Astrochronology of the end-Permian extinction and the Early–Middle Triassic. Goldschmidt 2018, Boston, USA. [Invited Oral]
34. **Mingsong Li**, Linda Hinnov, Chunju Huang, James Ogg. 2018. Sedimentary noise and Early Triassic sea levels linked to land–ocean water exchange and obliquity forcing. International Symposium on Deep-time Environmental & Climatic Extremes and Biotic Responses, Wuhan, China. [Invited Oral]
33. **Mingsong Li**, Linda Hinnov, Lee Kump. 2018. A time-series analysis software package (*Acycle*) for paleoclimate research and education. AGU Fall Meeting, Washington, DC, USA. [Poster]
32. Naihua Xue, Chengguo Guan, Bo Wang, **Mingsong Li**. 2018. Astronomical Time Scale of the Ediacaran Period of South China. AGU Fall Meeting, Washington, DC, USA. [Poster]
31. **Mingsong Li**, Lee R. Kump, Linda A. Hinnov, Michael E. Mann, Andy Ridgwell. 2018. New methods for astrochronological calibration of paleoclimate proxies and significance for deep-time data assimilation. The 3rd DeepMIP meeting, Bristol, UK. [Oral]

2017

30. **Mingsong Li**, Yang Zhang, Chunju Huang, James Ogg, Linda Hinnov, Yongdong Wang, Zhuoyan Zou, Liqin Li, Stephen Grasby, Yijiang Zhong, Keke Huang. 2017. Astrochronology and magnetostratigraphy of the Xujiahe Formation of South China and Newark Supergroup of North America: implications for the Late Triassic time scale. AGU Fall Meeting, New Orleans, USA. [Poster]
29. Lee Kump, **Mingsong Li** (presenter), Linda Hinnov. 2017. Tracking variable sedimentation rates in orbitally forced paleoclimate proxy series. AGU Fall Meeting, New Orleans, USA. [Poster]
28. **Mingsong Li**, Yang Zhang, James Ogg, Chunju Huang, Linda Hinnov, Zhong-Qiang Chen. 2017. Triassic time scale from astronomically tuned magnetostratigraphy. Geological Society of America Abstracts with Programs. 49(6), doi: 10.1130/abs/2017AM-302748. [Oral]

27. Yan Chen, Yang Zhang, James Ogg, **Mingsong Li**, Zhong-Qiang Chen, Chengyi Tu, Daoliang Chu. 2017. Magnetostratigraphy of the continental reference section for latest Permian through Early Triassic of North China at Dayulin (Henan province). Geological Society of America Abstracts with Programs. 49(6), doi: 10.1130/abs/2017AM-301904. [Oral]
26. **Mingsong Li**, Yang Zhang, James Ogg (presenter), Chunju Huang, Zhong-Qiang Chen, Linda Hinnov. 2017. Triassic time scale from astronomically tuned magnetostratigraphy. The 4th International Conference of Geobiology. Wuhan, China. [Oral]
25. **Mingsong Li**, Chunju Huang, James Ogg, Linda Hinnov, Yang Zhang, Zhong-Qiang Chen. 2017. Cyclo-magnetostratigraphy of the global Triassic. The First CUG Forum for Overseas Young Scholars. China University of Geosciences (Wuhan). [Oral]
24. Wei Liu, Huaichun Wu, Linda Hinnov, Chao Ma, **Mingsong Li**. 2017. Astronomically forced deposition in the Early Cretaceous Songliao synrift basin, China and its paleoclimatic implications. Joint 52nd Northeastern Annual Section / 51st North-Central Annual Section Meeting – 2017, Geological Society of America Abstracts with Programs. 49(2), doi: 10.1130/abs/2017NE-290749. [Oral]

2016

23. Linda Hinnov, **Mingsong Li**, Chunju Huang. 2016. Bio-cyclo-magneto-stratigraphy solves longstanding problems of the global Triassic: astronomically paced aquifer-eustasy, interregional platform correlation, and geologic time. AGU Fall Meeting, San Francisco, USA. [Invited Oral]
22. **Mingsong Li**, Chunju Huang, Linda Hinnov, James Ogg, Yang Zhang. 2016. On the time scale controversy of the Spathian substage (Early Triassic) in South China. Geological Society of America Annual Meeting, Denver, USA. [Poster]
21. **Mingsong Li**, Chunju Huang, Linda Hinnov, Weizhe Chen, Wei Tian. 2016. Astronomical-cycle scaling of the Anisian Stage in South China: implications for biotic recovery following the end-Permian mass extinction. Geological Society of America Annual Meeting, Denver, USA. [Oral]
20. Yang Zhang, **Mingsong Li**, James Ogg, Linda Hinnov, Chunju Huang, Zhong-Qiang Chen. 2016. Cycle-calibrated magnetostratigraphy and time scales for the Early and early-Late Triassic. International Geological Congress, Cape Town, South Africa. [Oral]

2015

19. **Mingsong Li**, Chunju Huang, Linda Hinnov. 2015. Testing multiple paleoclimatic proxies in a Triassic marine record from China. AGU Fall Meeting, San Francisco, USA. [Poster]
18. **Mingsong Li**, Chunju Huang, Linda Hinnov. 2015. Intermittent obliquity-forced climate during the Early Triassic. Geological Society of America Annual Meeting in Baltimore, MD, USA. [Oral]
17. **Mingsong Li**, Chunju Huang, Linda Hinnov, James Ogg, Yang Zhang, Zhong-Qiang Chen, Zhuoyan Zou. 2015. An astronomical time scale for Triassic ecosystem recovery in South China. Geological Society of America Annual Meeting in Baltimore, MD, USA. [Poster]
16. Yang Zhang, **Mingsong Li**, James Ogg, Paul Montgomery, Chunju Huang, Zhong-Qiang Chen, Zhiqiang Shi, Paul Enos, Daniel J. Lehrmann. 2015. Cycle-calibrated Magnetostratigraphy of middle Carnian from South China: Implications for Late Triassic Time Scale and Termination of the Yangtze Platform. Geological Society of America Annual Meeting in Baltimore, MD, USA. [Oral]
15. **Mingsong Li**, Linda Hinnov, Chunju Huang, James Ogg, Zhong-Qiang Chen, Yang Zhang. 2015. Cyclostratigraphy and Integrated Time Scale of the Early Triassic. 2nd Boreal Triassic Conference and 12th International Workshop on the Permo-Triassic, Svalbard, Norway. [Oral]
14. **Mingsong Li**, Chunju Huang, Linda Hinnov. 2015. Multiple proxies of ancient climate and sea-level change: a Permian–Triassic transition example. Geological Society of America Southeastern Section – 64th Annual Meeting. Chattanooga, Tennessee, USA. [Oral]

2014

13. **Mingsong Li**, James Ogg, Chunju Huang, Yang Zhang, Zhong-Qiang Chen, Linda Hinnov. 2014. Astrochronology of the Early Triassic from South China. IsoAstro Geochronology Workshop: The integration and intercalibration of radioisotopic and astrochronologic time scales, Madison, USA. [Oral]
12. **Mingsong Li**, Chunju Huang, James Ogg, Yang Zhang, Zhong-Qiang Chen, Linda Hinnov. 2014. Cyclostratigraphy of the Early Triassic from South China. The Third Conference on Earth Systems Science, Shanghai, China. [Oral]
11. **Mingsong Li**, Chunju Huang, Zhong-Qiang Chen, Yang Zhang, Linda Hinnov, James Ogg. 2014. Astrochronology of the Early Triassic from South China and the Earth system model. The 3rd International Conference of Geobiology – Combining ancient records with present-day observations. Wuhan, China. pp. 100–101. [Oral]

2013

10. **Mingsong Li**, Chunju Huang, Yang Zhang, Zhuoyan Zou, James Ogg. 2013. Astronomical calibration of the duration of the Induan Stage (Early Triassic). The 11th National Congress of the Palaeontological Society of China (PSC) and the 27th Annual Conference of PSC, Dongyang, China, p. 119. [Oral]
9. **Mingsong Li**, Chunju Huang, Yang Zhang, Zhuoyan Zou, Shifeng Tian, Xiaoqing Liu. 2013. High-resolution rock color series of the Permian–Triassic succession at the Xiakou section, Yichang, Hubei Province and its response to orbital forcing. Abstract of the 5th National Symposium on Sedimentology, Hangzhou, China, pp. 431–432. [Oral]
8. **Mingsong Li**, Chunju Huang, Yang Zhang, Shifeng Tian, Zhuoyan Zou, Xiaoqing Liu. 2013. Astrochronology of the Griesbachian substage (Lower Triassic) from the Xiakou section, Hubei Province, South China. In: Zhong-Qiang Chen, Hao Yang, Genming Luo (eds.), World Summit on P-Tr Mass Extinction & Extreme Climate Change, Wuhan, China, pp. 42–43. [Oral]
7. **Mingsong Li**, Yuewu Sun, Wenchun Ge, Xingzhou Zhang, Fengxu Zhang, Shuqin Zhang, Yanlong Zhang, Guowei Zhao, Dejun Zhang. 2013. Eastward termination of the Xar Moron River Suture in Jilin Province. The 6th National Symposium on Structural Geology & Geodynamics, Changchun, China, pp. 242–243. [Poster]

2010–2012

6. **Mingsong Li**, Yuewu Sun, Xingzhou Zhang, Wenchun Ge. 2012. Gaya River–Tumen River: site of closure of the Paleo-Asian Ocean. The Second Conference on Deep-Sea Research and Earth Systems Science, Shanghai, pp. 53–54. [Poster]
5. **Mingsong Li**, Yuewu Sun, Wenchun Ge, Yanlong Zhang. 2011. Age of the Kaishantun Flora in the Yanbian area, Jilin: constrained by detrital zircon U–Pb geochronology. 26th Annual Conference of Palaeontological Society of China, Guanling, pp. 99–100. [Poster]
4. Yuewu Sun, Xingzhou Zhang, **Mingsong Li**. 2011. New material on Permian phytogeography in the Yanbian area, eastern Jilin Province, China: Proceedings of Russian Science Conference with foreign participants on Geological processes in the lithospheric plate subduction, collision, and solid environments, Dalnauka, Vladivostok, 153. [Oral]
3. Yuewu Sun, Shuqin Zhang, **Mingsong Li**, Dejun Zhang. 2011. Carboniferous–Permian stratigraphy in the northeastern North China Plate. 26th Annual Conference of Palaeontological Society of China, Guanling, pp. 91–92. [Oral]
2. **Mingsong Li**, Yuewu Sun, Guowei Zhao. 2010. Permian phytogeography in the Yanbian area, Jilin. Annual Conference of Paleobotany sub-society of China, Jinghong, p. 32. [Oral]
1. **Mingsong Li**, Guowei Zhao, Chengwen Wang, Yuewu Sun. 2010. Eastern extension of the Xar Moron River Suture: evidence from Cathaysia flora from Daxinggou town, Wangqing County, NE

China. Symposium on Hydrocarbon Resources Investigation in Songliao Basin and adjacent area, Songyuan, pp. 69–70. [Oral]

8. Speaking Engagements

52. National Space Science Center, Chinese Academy of Sciences, Beijing, China, September 24, 2026. **Invited Speaker.**
51. Chongqing University of Science and Technology, Chongqing, China, May 13, 2026. **Invited Speaker.**
50. Peking University Library, Beijing, China, April 17, 2026. **Invited Speaker.**
49. Tongji University, Shanghai, China, January 26, 2026. **Invited Speaker.**
48. School of Atmospheric Sciences, Lanzhou University, Lanzhou, China, December 8, 2025. **Invited Speaker.**
47. School of Earth Sciences, Lanzhou University, Lanzhou, China, December 8, 2025. **Invited Speaker.**
46. Northwest University, Xi'an, China, December 1, 2025. **Invited Speaker.**
45. Shuangqing Forum, Beijing, China, November 24–25, 2025. **Invited Speaker.**
44. China University of Petroleum (East China), Qingdao, China, May 26, 2025. **Invited Speaker.**
43. Ocean University of China, Qingdao, China, May 27, 2025. **Invited Speaker.**
42. Tongji University, Shanghai, China, April 11–12, 2025.
41. Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences, Nanjing, China, January 14, 2025. **Invited Speaker.**
40. China University of Petroleum (Beijing), Beijing, China, January 2, 2025. **Invited Speaker.**
39. 7th Young Scholars Forum of Palaeontology, Chengdu, China, December 16, 2024. **Invited Speaker.**
38. Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China, December 4, 2024. **Invited Speaker.**
37. Yuanpei College, Peking University, Beijing, China, November 15, 2024. **Invited Speaker.**
36. Paris Observatory, Paris, France, July 17, 2024. **Invited Speaker.**
35. Workshop: Cyclostratigraphy Intercomparison Project 2.0, Brussels, Belgium, July 9, 2024. **Keynote Speaker.**
34. Shenyang Normal University, Shenyang, China, June 18, 2024. **Invited Speaker.**
33. School of Earth and Space Sciences, Peking University, Beijing, China, March 8, 2024. **Invited Speaker.**
32. Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, September 15, 2023. **Invited Speaker.**
31. Research Institute of Petroleum Exploration and Development, China National Petroleum Corporation, Beijing, China, April 8, 2023. **Invited Speaker.**
30. Institute of Tibetan Plateau Research, Chinese Academy of Sciences, Beijing, China, April 19, 2023. **Invited Speaker.**
29. Institute of Earth Environment, Chinese Academy of Sciences, “Legend of Past and Present” Forum, Beijing, China, April 13, 2023. **Invited Speaker.**
28. Paleoclimatology Group Seminar, Beijing, China, February 5, 2023. **Invited Speaker.**
27. Solid Earth Science Key Laboratory Alliance 2022 Academic Committee Meeting, Beijing, China, December 18, 2022. **Invited Speaker.**
26. National Natural Science Foundation of China, Continental Evolution and Monsoon System Evolution Basic Science Center Project, Online Academic Seminar Series, Beijing, China, October 15, 2022. **Invited Speaker.**
25. Jilin University, Changchun, China, May 9, 2022. **Invited Speaker.**
24. Cloud Meeting on Paleomagnetism, Beijing, China, December 5, 2021. **Invited Speaker.**
23. Peking University, Beijing, China, October 27, 2021. **Invited Speaker.**
22. Southern University of Science and Technology, Shenzhen, China, October 23, 2021. **Invited Speaker.**
21. Montclair State University, October 11, 2021. **Invited Speaker.**
20. Tongji University, Shanghai, China, October 11, 2021. **Invited Speaker.**
19. Peking University, Beijing, China, June 11, 2021.

18. Peking University, Beijing, China, May 26, 2021.
17. Tongji University, Shanghai, China, April 19, 2021. **Invited Speaker.**
16. China University of Geosciences (Beijing), Beijing, China, April 16, 2021. **Invited Speaker.**
15. Virginia Tech, Blacksburg, Virginia, USA, March 18, 2021. **Invited Speaker.**
14. Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China, March 16, 2021. **Invited LIU Tungsheng Lecture.**
13. Virginia Tech, Blacksburg, Virginia, USA, March 11, 2021. **Invited Speaker.**
12. Pennsylvania State University, State College, USA, February 14, 2020.
11. Lehigh University, Department of Earth and Environmental Sciences, Bethlehem, USA, September 5, 2019. **Invited Speaker.**
10. George Mason University, Department of Atmospheric, Oceanic and Earth Sciences, Geology Seminar Series, Fairfax, Virginia, USA, March 28, 2019. **Invited Speaker.**
9. Pennsylvania State University, State College, Pennsylvania, USA, September 21, 2018.
8. Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences, Nanjing, Jiangsu, China, May 11, 2018. **Invited Speaker.**
7. Nanjing University, Nanjing, Jiangsu, China, May 9, 2018. **Invited Speaker.**
6. China University of Geosciences, Wuhan, Hubei, China, March 26, 2018. **Invited Speaker.**
5. Purdue University, West Lafayette, Indiana, USA, January 18, 2018. **Invited Speaker.**
4. Peking University, Beijing, China, October 19, 2017. **Invited Speaker.**
3. Pennsylvania State University, State College, Pennsylvania, USA, April 14, 2017.
2. China University of Geosciences, Wuhan, Hubei, China, March 19, 2017. **Invited Speaker.**
1. George Mason University, Department of Atmospheric, Oceanic and Earth Sciences, Geology Seminar Series, Fairfax, Virginia, USA, October 8, 2015.

9. Teaching Experience

Courses

Fall 2026 Instructor, Peking University

01232010 Data Science for Geoscience (Undergraduate, Credits: 2)

Fall 2026 Instructor, Peking University

01217280 Seminar on Paleobiology and Stratigraphy (Graduate, Credits: 2)

Fall 2026 Instructor, Peking University

01201270 Deep-Time Earth Data Analysis and Visualization (in English, Graduate, Credits: 2)

Fall 2025 Instructor, Peking University

01232010 Data Science for Geoscience (Undergraduate, Credits: 2)

Fall 2025 Instructor, Peking University

01217280 Seminar on Paleobiology and Stratigraphy (Graduate, Credits: 2)

Fall 2025 Instructor, Peking University

01201270 Deep-Time Earth Data Analysis and Visualization (Graduate, Credits: 2)

Fall 2024 Instructor, Peking University

01232010 Data Science for Geoscience (Undergraduate, Credits: 2)

Fall 2024 Instructor, Peking University

01217280 Seminar on Paleobiology and Stratigraphy (Graduate, Credits: 2)

Fall 2024 Instructor, Peking University

01201270 Deep-Time Earth Data Analysis and Visualization (Graduate, Credits: 2)

Fall 2023 Instructor, Peking University

01232010 Data Science for Geoscience (Undergraduate, Credits: 2)

Fall 2023 Instructor, Peking University

01201270 Deep-Time Earth Data Analysis and Visualization (Graduate, Credits: 2)

Fall 2022 Instructor, Peking University

01201270 Deep-Time Earth Data Analysis and Visualization (Graduate, Credits: 2)

Spring 2021, Spring 2022 Instructor, Peking University*Advances in Paleontology and Stratigraphy* (Graduate, Credits: 2, co-taught)**Fall 2019 Interim Instructor, Pennsylvania State University***GEOSC 204 – Geobiology*: Co-designed and taught lectures on paleoclimate and mass extinctions.**September 2019 Invited Short Course, Lehigh University***Acycle*: Time-series analysis software for paleoclimate research and education.**Spring 2016 Interim Instructor, George Mason University***GEOL 304 – Sedimentary Geology: Analysis and interpretation of sediment, sedimentary rocks, and strata*. Taught 10+ students to perform chronostratigraphic analysis using MATLAB and Excel to understand cyclical patterns of climate change in the Newark Basin.**Spring 2016 Interim Instructor, George Mason University***GEOL 565 / CLIM 759 – Paleooceanography: Investigation of ocean evolution through geologic time*. Taught 10+ students the basics of geologic time.**2014 Teaching Assistant, China University of Geosciences (Wuhan)***Astronomical Forcing of the Paleoclimate System*: Guided 25+ students in understanding astronomical forcing of the climate and cyclostratigraphy via hands-on astrochronological lab work and the interpretation of chronostratigraphic information.**Workshop Organization**April 2025, Acycle Cyclostratigraphy Workshop, Beijing, China. **Chair** (400+ participants).**Short Courses**

April 2025, Short Course on Cyclostratigraphy, Sedimentological Congress, Beijing, China.

Organizer and Lecturer.

April 2023, Short Course on Cyclostratigraphy, Sedimentological Congress, Chengdu University of Technology, Chengdu, China. Lecturer.

OtherJuly 2026, AI-Empowered Research and Education, Peking University, Beijing, China. **Invited Speaker**.April 2026, The Logical Structure of High-Level Scientific Research: From Problem Awareness to Communicating Findings, Higher Education Press, Beijing, China. **Invited Speaker**.January 2026, “Research Paper Writing in the AI Era”, Wuhan Business University, Wuhu, Anhui, China. **Invited Speaker**.December 2024, Generative AI Revolutionizing Geoscience Research and Education, AI for Education Yanyuan Zhisheng Salon, Peking University, Beijing, China. **Invited Speaker**.

10. Student and Postdoctoral Mentorship

Postdoctoral Researchers

Haotian Zhang: Boya Postdoctoral Fellow, since July 2026; supervisor

Graduate Students

Haotian Zhang: Ph.D. student, 2021–2026; supervisor

Xiaoyu Zhang: Ph.D. student, since September 2022; supervisor

Hanyu Zhu: Ph.D. student, since September 2023; supervisor

Zhixin Wang: Ph.D. student, since September 2024; supervisor

Xinwen Zhang: Ph.D. student, since September 2025; supervisor

Jiahao Liu: Master's student, since September 2026; supervisor

Undergraduate Thesis Supervisor

2023 Hanyu Zhu

2024 Zhixin Wang
 2025 Xinwen Zhang
 2026 Jiahao Liu
 2027 Heng Wu (expected)
 2027 Jiayuan Ren (expected)
 2028 Rumeng Liu (expected)
 2029 Zhenglei Wu (expected)

Past Lab Members

Meng Wang: 2021–2023, Peking University Boya Postdoctoral Fellow; 2019–2020, visiting Ph.D. student at Penn State; currently an Associate Research Professor, Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences (supported by the CAS Hundred Talents Program, Category B)
 Kaixuan Ji: 2022–2026, Postdoctoral Researcher; currently a Lecturer, China University of Geosciences (Beijing)
 Yujing Wu: 2021–2023, Ph.D. degree, associate supervisor; currently a postdoctoral researcher at IMCCE, Observatoire de Paris (advisor: Jacques Laskar)
 Qingqing Jiang: Master's student, 2022–2025; supervisor; currently a Ph.D. student at the University of Münster
 Haoxun Zhang: Master's student, 2021–2024; supervisor
 Ciro Clímaco Rodrigues: 2021–2023, master's degree, National Observatory in Rio de Janeiro, Brazil; associate supervisor; currently at Petrobras
 Mariana Aragão Fernandes: 2021–2023, master's degree, National Observatory in Rio de Janeiro, Brazil; associate supervisor

Undergraduate Class Adviser

Geochemistry Strong Foundation Program, Class 1, Cohort 2022, School of Earth and Space Sciences, Peking University (recognized as an “Outstanding Class Collective” of the School for the 2022–2023 academic year)

11. Projects

Since 2021: RMB 10,519,300 in total funding (USD 1.53 million)

8. January 2025 – December 2027, National Natural Science Foundation of China, Original Exploration Program, “Revealing Carbon Release Sources and Processes of the Paleocene–Eocene Thermal Maximum through Cross-Sphere Isotope Data Assimilation (Main Application)” (42450195), **PI**, RMB 1,741,500
7. January 2025 – December 2028, National Natural Science Foundation of China, “Cyclostratigraphy of the secular resonance transition event of planetary orbits in the early Cenozoic chaotic Solar System” (42472050), **PI**, RMB 689,000
6. December 2022 – November 2027, National Key R&D Program of China (2022YFF0802900), “Temporal and spatial evolution of orbital scale ocean deoxygenation during the Paleocene–Eocene Thermal Maximum”, **PI**, RMB 4,000,000
5. November 2021 – October 2026, National Key R&D Program of China (2021YFA0718200), “Metal stable isotope geochemical technique for tracing the evolution of the Earth's habitability in the middle Proterozoic”, **Co-PI**, RMB 1,333,300
4. January 2022 – December 2025, National Natural Science Foundation of China, “Astrochronology and past global change”, **PI**, RMB 1,000,000
3. January 2021 – December 2024, National Natural Science Foundation of China (No. 42072040), “Study on the mechanism of lake level and sea level changes in the Early Triassic of China and Germany based on sedimentary noise modeling”, **PI**, RMB 725,500
2. January 2021 – December 2022, the Fundamental Research Funds for the Central Universities (No. 7100603368), “High resolution astrochronology and paleoclimate change”, **PI**, RMB 1,000,000

1. July 2021 – June 2022, Hubei Key Laboratory of Critical Zone Evolution, China University of Geosciences, Wuhan (No. 2021F07), “Sea level reconstruction from the Lower Triassic Xiejiacao section of Guangan, Sichuan”, **PI**, RMB 30,000

12. Professional Activities

(1) Committee

- 2025–2030, Vice Chairman, Committee on Palaeogeographic Reconstruction Technology and Big Data, International Society of Palaeogeography
 2022–present, Committee Member, Anthropocene Research Society, Geological Society of China

(2) Editorial Board

- 2021–2027, *National Science Review*, Editorial Group Member
 2026–2027, *Science Bulletin*, Youth Editorial Board Member
 2025–present, *Continent and Life Evolution*, Youth Editorial Board Member
 2024–present, *Acta Sedimentologica Sinica*, Youth Editorial Board Member
 2023, *Quaternary Sciences*, Invited Editorial Member

(3) Conference Chair and Convener

- 2027 European Geosciences Union (EGU) General Assembly, Vienna, Austria, **Co-Convener**, Session *CL4.1: Orbital forcing of global and regional events throughout Earth history*
 2026 The 32nd Annual Meeting of the Palaeontological Society of China (PSC), Xi'an, China, **Convener**, *S19: High-Resolution Integrated Stratigraphy and Geochronology*
 2026 The 7th International Palaeogeography Conference, Mendoza, Argentina, **Convener**, *T7-7: Astronomical forcing of palaeoclimate and palaeoenvironmental systems*
 2026 The 5th International Stratigraphic Congress (STRATI), Kunshan, Jiangsu, China, **Co-Convener**, *G12: Cyclostratigraphy and its applications in geochronology and paleoclimatology*
 2026 European Geosciences Union (EGU) General Assembly, Vienna, Austria, **Co-Convener**, Session *CL4.1: Orbital forcing of global and regional events throughout Earth history*
 2026 The 9th Chinese Conference on Artificial Intelligence and Big Data for Earth Sciences & The 4th Summit Forum on Spatial Earth Big Data, Hefei, Anhui, China, **Co-Convener**, Topic: *Big Data, Artificial Intelligence, and the Evolution of Habitable/Early Earth*
 2025 The 3rd National Symposium on Data-Driven Earth Science Development, Zhuhai, Guangdong, China, **Co-Convener**, Topic *24: AI Paradigms in Big Data Earth Science Research*
 2025 Conference on Earth System Science (Shanghai), **Convener**, *Basin Sedimentary Systems and the Evolution of the Global Water Cycle*
 2025 Conference on Earth System Science (Shanghai), **Co-Convener**, *Paleocean and Paleoclimate of the Greenhouse Earth*
 2025 Conference on Earth System Science (Shanghai), **Co-Convener**, *Pre-Cenozoic Orbital Forcing and Global Change*
 2025 8th National Sedimentology Conference, **Co-Convener**, *High-Resolution Chronostratigraphy of Fine-Grained Sediments*
 2025 8th National Sedimentology Conference, **Co-Convener**, *Orbitally Driven Environmental and Climate Evolution*
 2024 AGU Fall Meeting, **Co-Chair**, *PP53G: Astronomical Forcing of Earth's Paleoclimate System II (Oral)*
 2024 AGU Fall Meeting, **Co-Chair**, *PP51B: Astronomical Forcing of Earth's Paleoclimate System I (Poster)*
 2024 2nd National Symposium on Data-Driven Geosciences, **Co-Convener**, Session *1: Big Data and Artificial Intelligence Paradigm in Earth Science Research*

- 2024 The 6th International Conference of Palaeogeography, **Co-Chair**, *T7-8: Astronomical Forcing of Paleoclimate and Paleoenvironmental Systems*
- 2023 AGU Fall Meeting, **Co-Chair**, *PP01: Online Poster Session for Paleooceanography and Paleoclimatology I*
- 2023 AGU Fall Meeting, **Chair and Primary Convener**, *PP21D: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System I Poster*
- 2023 AGU Fall Meeting, **Chair and Primary Convener**, *PP24B: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System II Oral*
- 2023 Conference on Earth System Science (Shanghai), **Chair and Convener**, Session 36: Long-period orbital forcing of Earth system evolution
- 2023 Conference on Earth System Science (Shanghai), **Chair and Convener**, Session 37: Paleoenvironmental reconstruction during periods of global warming
- 2022 AGU Fall Meeting, **Chair and Primary Convener**, *PP11C: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System I Online Poster Discussion*
- 2022 AGU Fall Meeting, **Chair and Primary Convener**, *PP13B: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System II Oral*
- 2022 AGU Fall Meeting, **Chair and Primary Convener**, *PP25D: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System III Poster*
- 2022 AGU Fall Meeting, **Co-Chair and Co-Convener**, *PP14A: Ocean Deoxygenation During Past Hyperthermals I Online Poster Discussion*
- 2022 AGU Fall Meeting, **Co-Chair and Co-Convener**, *PP22D: Ocean Deoxygenation During Past Hyperthermals II Poster*
- 2021 AGU Fall Meeting, **Primary Convener**, *PP006: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System*
- 2021 GSA Annual Meeting, **Co-Convener**, *T171: Data-Driven Approaches Deciphering Water and Carbon Cycles in Earth-Surface Systems*
- 2021 Virtual European Geosciences Union General Assembly, **Co-Convener**, *CL1.8/SSP2.4: Climate response to orbital forcing*
- 2020 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Astronomical Forcing and Past Climate Cycles*
- 2020 Goldschmidt Conference, **Co-Chair and Co-Convener**, *Topical Session: New Developments in Deep-Time Paleooceanography: Geochemical Proxies, Cyclostratigraphy & Data Analysis*
- 2019 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Cyclostratigraphy and Astronomical Forcing of Past Climates*
- 2019 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Chrono-stratigraphy using Magnetic Methods*
- 2018 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Cyclostratigraphy and Astrochronology in Deep Time*
- 2018 International Symposium on Deep-time Environmental & Climatic Extremes and Biotic Responses, Wuhan, China, **Co-Chair**, *Session: Triassic*
- 2018 European Geosciences Union General Assembly, **Co-Convener**, *Session CL1.31: Climate response to orbital forcing*
- 2017 AGU Fall Meeting, **Co-Chair and Co-Convener**, *Topical Session PP42B: Cyclostratigraphy and Astronomical Forcing of Past Climates*
- 2017 GSA Annual Meeting, **Co-Chair and Co-Convener**, *Topical Session T47: Recent Developments in Cyclostratigraphy*

(4) Proposal Referee

China, National Natural Science Foundation of China (NSFC)

USA, National Science Foundation (NSF)

UK, Natural Environment Research Council (NERC)

Germany, Deutsche Forschungsgemeinschaft (German Research Foundation, DFG)

Poland, Polish National Science Centre (NCN)

Netherlands, Dutch Research Council (NWO)

(5) Journal Referee (130+ review reports)

Publons: <https://publons.com/researcher/1455928/mingsong-li/peer-review/>

(1) Science, (1) Science Advances, (3) Nature Communications, (3) PNAS, (31) Palaeogeography, Palaeoclimatology, Palaeoecology, (11) Global and Planetary Change, (8) Paleoceanography and Paleoclimatology, (7) Earth and Planetary Science Letters, (7) Earth-Science Reviews, (5) Journal of Asian Earth Sciences, (4) GSA Bulletin, (4) Marine and Petroleum Geology, (3) Climate of the Past, (3) Computers & Geosciences, (3) Cretaceous Research, (3) Geological Journal, (3) Geology, (3) Geophysical Research Letters, (2) Scientific Reports, (3) Quaternary Science Reviews, (3) Science China Earth Sciences, (2) Gondwana Research, (2) Journal of King Saud University - Science, (2) Science Bulletin, (2) Terra Nova, (1) Arabian Journal of Geosciences, (1) Frontiers in Earth Science, (1) Geoscience Frontiers, (1) Journal of Earth Science, (1) Journal of Geophysical Research: Atmospheres, (1) Journal of Ocean University of China, (1) Journal of the Geological Society, (1) MethodsX, (1) Paleoceanography, (1) Precambrian Research, (1) Sedimentology

(6) Judge

2022, 5th “Academic Star” Competition for Graduate Students, School of Earth and Space Sciences, Peking University

2021, 4th “Academic Star” Competition for Graduate Students, School of Earth and Space Sciences, Peking University

2019 Outstanding Student Presentation Award (OSPA), AGU

2018 Outstanding Student Presentation Award (OSPA), AGU

2017 10th Annual Postdoc Research Exhibition, Penn State

(7) Science Communication

Acycle Time-Series Analysis WeChat Official Account (Owner, approximately 7,000 subscribers)

13. Selected Media Coverage

2026 Peking University Homepage, Peking University News, and the International Organization Affairs Platform of the China Association for International Science and Technology Cooperation. “Mingsong Li of the School of Earth and Space Sciences, Peking University, Receives the 4th International Sun Shu Award” / “Mingsong Li of Peking University Wins an International Award!”

<https://news.pku.edu.cn/xwzh/dda45c73785b41b78522f75364203159.htm>

2026 *National Science Review*, *Intellectual*, Tencent News, and Sina Finance. “Exploring the Chaotic Future and the Deterministic Past of the Solar System: An Interview with Jacques Laskar” / “How Long Can the Stability of the Solar System Last?”

<https://academic.oup.com/nsr/advance-article/doi/10.1093/nsr/nwag002/8414053>

2026 *ScienceNet (China Science Daily)* and Guangzhou Institute of Geochemistry, Chinese Academy of Sciences. “Scientists Precisely Constrain the Duration of a Key Geomagnetic Reversal Event” / “Lu Guimei et al.-SA: Duration of M0r and OAE1a and Its Implications for the Global Carbon Cycle”

<https://news.sciencenet.cn/htmlnews/2026/3/561005.shtm>

2025 Peking University Homepage and Peking University News. “Schools of Earth and Space Sciences and Mathematical Sciences at Peking University Awarded NSFC Original Exploration Program Project”

<https://news.pku.edu.cn/xwzh/9a4af42b70c34174bb7072d9af29d0a8.htm>

- 2025 *Technology Networks*. “Ancient Ocean Acidification Parallels Modern Climate Change”
<https://www.technologynetworks.com/applied-sciences/news/ocean-acidification-during-the-paleocene-offers-parallels-to-modern-climate-change-394586>
- 2025 Science China Press and EurekAlert! (English and Chinese editions). “Unlocking Earth’s Past: How Data Assimilation Is Revolutionizing Paleoclimate Research” / “解密地球过去：数据同化如何革新古气候学研究”
<https://www.eurekalert.org/news-releases/1084549>
- 2025 Peking University News (Chinese and English editions). “Acycle Cyclostratigraphy Workshop Held at Peking University”
<https://news.pku.edu.cn/xwzh/b8ace2a66b654b2798e5f9698b335ab1.htm>
- 2025 Peking University Homepage, Peking University News, School of Ocean and Earth Science, Tongji University, State Key Laboratory of Marine Geology, Tongji University, Sohu “Marine Knowledge Circle,” and Center for Paleontology and Stratigraphy, Jilin University. “Three Peking University Scholars Receive the 4th China Sun Shu Award” / “The 4th Young Sedimentologists Forum Successfully Held at Tongji University”
<https://news.pku.edu.cn/xwzh/73b5704264d34ec6aead8404402d1a36.htm>
- 2024 Digital Intelligence Platform for Arts and Humanities, Peking University / Peking University News. “AI for Education Yanyuan Zhisheng Salon: Educational Transformation and Strategies in the Era of Generative AI (Episode 5)”
<https://news.pku.edu.cn/xwzh/c6eedddc69944c27b7696ad6b0a1b3cb.htm>
- 2024 Xinhua News Agency (English). “Ancient Climate Crisis Offers Warning on Modern Ocean Acidification: Study”
https://xhnewsapi.xinhuaxmt.com/share/news_pc?id=1030777941192704
- 2024 CGTN, China Economic Net, United News of Bangladesh, TBSNews.net. Ancient climate crisis offers warning on modern ocean acidification <https://news.cgtn.com/news/2024-12-23/Ancient-climate-crisis-offers-warning-on-modern-ocean-acidification-1zyJSjVik9y/p.html>
- 2024 Phys.org. Ancient climate study links past ocean acidification to current trends
<https://phys.org/news/2024-11-ancient-climate-links-ocean-acidification.html>
- 2024 noticiasdelatierra. Un estudio sobre el clima antiguo vincula la acidificación pasada de los océanos con las tendencias actuales <https://noticiasdelatierra.com/un-estudio-sobre-el-clima-antiguo-vincula-la-acidificacion-pasada-de-los-oceanos-con-las-tendencias-actuales/>
- 2024 Ocean Acidification. Ancient climate study links past ocean acidification to current trends
<https://news-oceanacidification-icc.org/2024/11/20/ancient-climate-study-links-past-ocean-acidification-to-current-trends/>
- 2024 Ocean Acidification. Coupled decline in ocean pH and carbonate saturation during the Palaeocene–Eocene Thermal Maximum <https://news-oceanacidification-icc.org/2024/11/20/coupled-decline-in-ocean-ph-and-carbonate-saturation-during-the-palaeocene-eocene-thermal-maximum/>
- 2024 北京大学官网首页, 北京大学新闻网, 地空学院李明松团队揭示远古气候危机时期的海洋酸化与当今的相似之处 <https://news.pku.edu.cn/jxky/873f4e2126a24e0b8100a21bb60c6c31.htm>
- 2024 北京大学地空学院, 北大地空学院李明松揭示远古气候危机时期的海洋酸化与当今的相似之处 <https://sess.pku.edu.cn/info/1291/4648.htm>
- 2024 微信公众号(Acycle 时间序列分析, 中国大洋钻探计划, 地学之家, 化学星球, Marine Sedimentology 等), 远古大洋的挽歌：温室气体排放、海洋酸化与生物大灭绝
https://mp.weixin.qq.com/s/oSJQSGCQC_PUdKMDocWOW

- 2023 北京大学新闻网, 国家重点研发计划“地球系统与全球变化”重点专项青年项目启动会
<https://news.pku.edu.cn/xwzh/89b5392190ed4092b66a61ac0de7e1df.htm>
- 2023 北京大学地空学院, 国家重点研发计划“地球系统与全球变化”重点专项青年项目启动会
<https://sess.pku.edu.cn/info/1033/3941.htm>
- 2022 北京大学新闻网, 地空学院李明松研究员在古新世-始新世极热事件研究中取得进展,
<https://news.pku.edu.cn/jxky/20c6bb15765e489ba497b0ed382b38e7.htm>
- 2022 Penn State, ScienceDaily, Phys.org: Changes in Earth's Orbit May Have Triggered Ancient Warming Event <https://www.psu.edu/news/research/story/changes-earths-orbit-may-have-triggered-ancient-warming-event>
- 2022 Episode hyperthermique du Paléocène-Éocène : et si la cause était astronomique ?
<https://sciencepost.fr/episode-hyperthermique-du-paleocene-eocene-et-si-la-cause-etait-astronomique/>
- 2022 LOS CAMBIOS EN LA ÓRBITA DE LA TIERRA PUEDEN HABER DESENCADENADO UN ANTIGUO EVENTO DE CALENTAMIENTO <https://noticiasdelatierra.com/los-cambios-en-la-orbita-de-la-tierra-pueden-haber-desencadenado-un-antiguo-evento-de-calemtamiento/>
- 2022 Une modification dans l'orbite de la Terre a causé le plus grand réchauffement climatique jamais connu <https://www.futura-sciences.com/planete/actualites/rechauffement-climatique-modification-orbite-terre-cause-plus-grand-rechauffement-climatique-jamais-connu-102342/#xtor%3DRSS-8>
- 2022 Ученые выяснили последствия изменений орбиты Земли
<https://cursorinfo.co.il/interest/uchenye-vyyasnili-posledstviya-izmenenij-orbity-zemli/>
- 2022 Выявлены причины древнего глобального потепления
<https://lenta.ru/news/2022/12/15/orbital/>
- 2022 Maps of the past may shed light on our climate future <https://phys.org/news/2022-10-climate-future.html>
- 2022 中国科学报, 学术论文中, 署名应该遵守什么规范,
<https://news.sciencenet.cn/htmlnews/2022/5/479615.shtm>
- 2021 The Paper, PKU News, Phys.org, Courthouse News Service, “Volcanic eruptions that caused Permian mass extinction also brought huge spike in global temperatures”
<https://phys.org/news/2021-09-mass-extinction-lethal-temperatures-due.html>
- 2021 lenta.ru, Названа причина крупнейшей катастрофы в истории Земли,
<https://lenta.ru/news/2021/09/09/traps/>
- 2021 MSN, Yahoo, Le CO2 en cause dans la pire catastrophe de l'Histoire de la Terre,
<https://www.msn.com/fr-fr/actualite/technologie-et-sciences/le-co2-en-cause-dans-la-pire-catastrophe-de-lhistoire-de-la-terre/ar-AAOcuZ8?li=BB0JvSH>, <https://fr.news.yahoo.com/co2-cause-pire-catastrophe-l-014500489.html>
- 2021 nrc.nl, Siberische vulkanen zorgden voor massaal uitsterven,
<https://www.nrc.nl/nieuws/2021/09/06/siberische-vulkanen-zorgden-voor-massaal-uitsterven-a4057312>
- 2021 澎湃新闻、科学网, “史上最大物种灭绝罪魁祸首找到了：火山二氧化碳，海水被酸化”
https://www.thepaper.cn/newsDetail_forward_14391612,
<http://news.sciencenet.cn/htmlnews/2021/9/464658.shtm>
- March 2020 Newsweek: “Days on Earth When Dinosaurs Lived Were Half an Hour Shorter Than They Are Now, Ancient Fossil Reveals” <https://www.newsweek.com/days-earth-dinosaurs-half-hour-shorter-ancient-fossil-1491422>
- October 2019 Penn State Today, ScienceDaily, Physics News: “Ancient rain gauge: New evidence links groundwater, climate changes in deep time”
<https://phys.org/news/2019-11-ancient-gauge-evidence-links-groundwater.html>

- April 2018 Penn State Today, ScienceDaily, Physics News, and EurekAlert!: *“Connection of sea level and groundwater missing link in climate response”*
<https://www.sciencedaily.com/releases/2018/04/180403120002.htm>
- April 2018 Europa Press: *“La inclinación terrestre altera el nivel del mar sin hielo en los polos”*
<http://www.europapress.es/ciencia/habitat-y-clima/noticia-inclinacion-terrestre-altera-nivel-mar-hielo-polos-20180403143840.html>
- September 2017 Chinese Academy of Sciences: *“A Step Toward A Complete Triassic Time Scale: Proposal from China”*
http://english.cas.cn/newsroom/research_news/201709/t20170905_182756.shtml
- September 2017 Xinhua News Agency: *“New evidence for the Geological Time Scale from China”*
http://news.xinhuanet.com/tech/2017-09/06/c_1121616041.htm
- September 2017 People.cn and China University of Geosciences (Wuhan): *“Scientists make progress in the area of the Triassic timescale”*
<http://hb.people.com.cn/n2/2017/0903/c337099-30687078.html>
<http://www.cug.edu.cn/info/10506/88848.htm>
- July 2016 Science Net: *“Researchers of the China University of Geosciences: Astronomically forced catastrophes during the Early Triassic”*
http://news.sciencenet.cn/dz/dznews_photo.aspx?id=25812
- July 2016 People.cn: *“Status of Earth's time: 22-hour length-of-day and a 398-day year 250 million years ago”*
<http://hb.people.com.cn/n2/2016/0702/c337099-28599380.html>